Abstract - Integrating the latest technology such as blockchain, cloud computing, and artificial intelligence, the FinTech industry has grown and evolved in a short period. Nevertheless, due to this very speed of development, the financial services have become more vulnerable to cybersecurity threats, including such advanced malware attacks as those that led to the loss of billions of dollars. This report involves the analysis of the effectiveness of the existing cybersecurity defences in the FinTech industry to propose innovative ideas of making security more resilient. It demonstrates the two challenges that the financial institutions face in the struggle to advance their level of technology and preserve finances against escalating risks relating to cyberspace assuming threats through the internet persons. This paper aims at developing and implementing an improved intrusion detection system (IDS) considering the needs of the FinTech industry based on the use of deep neural networks. Such a strategy would significantly increase the level at which cybersecurity threats are detected, as it would minimize the occurrence of false positives and false negatives, two of the main issues associated with the current security systems. The proposed IDS can improve the whole security system of financial firms, and it is possible to identify the threats more accurately reducing such errors. The report equally has an elaborate defence plan to solve emerging and increasing security risks post-2020. The method summarizes a blend of implementing innovative cybersecurity tools and carefully reviewing the prospective future threats to enhance the functional matters. efficient and robustness of the financial institutions.

Its aim is to develop an effective defence mechanism that will avert existing identify cybersecurity threats and predict upcoming ones and protect the sustainability and integrity of FinTech services amidst the greater digitalization of the economy. To put it differently, this strategy underscores how much it matters that cybersecurity policies should be constantly innovative to follow the new technologies and the never-stagnating realm of cyberthreats.

Key Term

FinTech, Malware, Worm, Ransomware, Spyware, Cyber Security, Ddos, Intrusion Detection System (IDS), Deep Neural Network (DNN).

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1. Introduction

The review discusses the evolving sophistication and considerable intensity of malware risks that are affecting the financial technology (FinTech) industry, especially with the increasing cases of malware being encountered in the financial technology nature of business. The appearance of the opportunities provided by developing technologies, such as artificial intelligence (AI), neural networks, cloud computing, blockchain, and embedded systems, has caused the dramatic change in the FinTech. As a result of this revolution, FinTech has become a critical component of the world

financial system and contributed to automation of financial services as well as designing innovative business approaches. Javaheri et al. In the year 2023, a cross-platform malware is working out and that may encroach into several FinTech platforms. Over 40 percent of all cyberattacks on banking and insurance organizations around the world occurred over the last 12 months (October 2021 to September 2022) due to malware. A total of 22 million devices and 721.5 million credentials had been stolen by cybercriminals. In 2022, 39 percent of all corporate FinTech users and 61 percent of all consumers worldwide were being targeted by cybercriminals via financial malware attacks. The Central Bank of Bangladesh had been hacked by cybercriminals in February 2016 using APT software. By using the compromised credentials to access the SWIFT system, they were able to submit a request that resulted in the conversion of 81 million US dollars from the Central Bank of Bangladesh's foreign currency reserves to the Federal Reserve Bank of New York. When more than US$2 million was taken from the City Union Bank of India and other South American institutions in 2018, hackers used the identical technique used by the Central Bank of Bangladesh. Equally, there was a case by Taiwanese programmers who stole over 70 million new Taiwan money through teller machines by use of a software. In order to create backdoors that allow hackers to access sensitive data (passwords, usernames, and credit card numbers) and circumvent authentication and verification processes, hackers usually install and disseminate malware on mobile devices and within the network after breaching the victims' computer systems and networks. Ransomware and Trojan horse are the two types of malware commonly used in the banking industry (Ali et al., 2024). 2021 turned out to be a year of reckoning, with losses as high as $1 billion documented in the literature on global cybercrime. Financial institutions, particularly Belief Networks (DBNs) and its variants, were the main target of the heist (Kamuangu, 2024). These vulnerabilities make data and financial transactions insecure, which makes the industry an attractive target for hackers. The broad objectives of this study to increase the effectiveness and prediction accuracy of IDS demonstrate its creative approach.

The system has the capacity to adapt to constantly shifting landscape of cyber threats by integrating such aspects as reduction in feature space, modification of the importance of classes, and data distribution optimisation between training and test data. This is to ensure that IDS has flexibility to respond to new threats so that the fintech market has an effective protection system. By keeping malware in mind, as it has

become tremendously complex and prevalent and is now even affecting the financial system negatively, this paper tries to establish the intricacies that are involved during this digital conflict.

Large rates of false negatives and false positives tend to hinder the current remedies. As Javaheri et al. (2023) note, in 2023, the funding received was over 5.9 billion. In the case of the FinTech industry, the proposed research recommends an intrusion detection system (IDS) with deep neural networks. In enhancing the precision in the actual risks, this would reinforce security architecture of the financial institutions and reduce unnecessary operational interference (Thakkar & Lohiya, 2023). This intellectual endeavour aims at providing an effective solution to financial organisations.

1.1 Problem Statement

Malware is one of the greatest threats to cyber space. In order to make this better, the Financial Services segment was the most preferred by malware as reported by Evolve taping 19 per cent of all malware assaults with over 18.3 billion dollars loss recorded back in 2017 alone. The sectors that were affected to the next degree were the Utilities and Energy. Cybercrime in the FinTech sector is defined as fraud activities, conducted through mobile devices and other networking technologies to destroy technology, embarrass the clients, cause monetary loss, steal confidential information and cause panic and concern. Cybersecurity data indicates an average of 2,200 cyber attacks every 39 seconds indicating that by 2027, cybercrime will healthcare cost 23.82 trillion dollars. Hackers get the capability to crack and shut down servers, mobile phones and computer networks due to malware; encrypt, delete, modify, spy, and take away personal data and registration files; and even obtain access to finance networks and apps without permission. Malware is evolving and may find its way into a system by seducing the authorised users into opening malicious files, browsing malicious websites or download and install malware unintentionally through obtaining access to sources where its installation has been done (Ali et al., 2024). The use of machine learning-based intrusion detection systems (IDS), particularly to monitor Low-Rate Distributed Denial of Service (LR-DDoS) attacks, is one of the possible solutions. To detect such attacks with high accuracy, these systems employ tree-based classifiers, such as Random Forest (RF) and Support Vector Machine (SVM) (Javaheri et al., 2023).

Nevertheless, this current solution can experience low false -positive ratio or the false- negative ratio. This is because there are chances that the datasets of attacks generated have a lot of excess and redundant data thus making intrusion detection systems (IDS) who are based on ML useless in detecting a diverse set of threats. Therefore, such a solution to various forms of network attack and its variations as the creation of a smart and successful mechanism like a deep neural network (Thakkar & Lohiya, 2023).

1.2 Project Aim

The aim of this study is to propose a refined taxonomy of cybersecurity threats exists in FinTech industry and this authoritative knowledge would be of use in evaluating the effectiveness of the current defensive methods to provide a systematic and comprehensive understanding of threats.

1.3 Project Objectives

1) To identify and categorize most relevant and upcoming cybersecurity threats in FinTech.
2) To evaluate and analyze the efficiency of the available defensive policies used by financial institutions.
3) To propose and reforms to the existing counter-defensive measures that will allow them to deal more effectively with the changes and complexities of the emerging cyber threats.

1.4 Project Scope (Tasks, Constraints, and Deliverables)

1) Tasks need to be done
   - Task 1: Data Acquisition and Exploratory Data Analysis (EDA)

The first stage would be to find and secure appropriate, publicly accessible and anonymous transaction data. It will then be followed by profound exploratory data analysis (EDA) to realize the structure of the data, its important features, and distribution of fraudulent and authentic transactions. The task will be completed with

the visualizations of the transaction regularities, the correlation between the features, and the extreme class imbalance peculiar to the data set.

- Task 2: Data Preprocessing and Feature Engineering

This task expands on EDA and is about the application of sound data cleaning and preprocessing procedures, such as missing value handling, feature scaling, and feature encoding. Raw data will result in new, relevant features (e.g. creating time based features, calculating the transaction frequency of each user or obtaining speed based features). Lastly, the data preparation approach will be carried out on class imbalances using oversampling (SMOTE) or under sampling techniques to complete the process of training an effective model.

- Task 3: Model Implementation and Training

This involves the central development of fraud detection model. A range of baseline machine learning models (logistic regression, random forests and gradient boosting models (XGBoost)) will be applied and trained. At least one advanced deep learning model will also be designed, implemented and trained in the project; it may be a single or more architectures, with the primary emphasis on one architecture, such as autoencoders to capture anomalies or recurrent neural networks (RNN/LSTM) to capture time dependencies. Each model is going to be tuned with the help of Hyperparameter tuning.

- Task 4: Model Evaluation and Comparison

Determining the developed model will be done through a range of metrics, which are appropriate to deal with imbalanced data, such as accuracy, recall, F1 score, and ROC-AUC score. A confusion matrix will be constructed and investigated to realize the kind of errors that the models will make (false positive errors and false negative errors). Lastly, comparison will be done between the traditional machine learning models and the deep learning models to identify the pros and the cons of both of the models.

- Task 5: Reporting and Documentation

The last activity is when the author will prepare a project report that includes the complete process such as literature investigation, methods, findings, discussion, and conclusion. There will be complete documentation of the source code including comments and explanations so that there can be rigor and reproducibility.

2) Constraints

- Time Constraint

The projects have to be completed within the allocated semester and this restricts the level of depth and complexity in some tasks like broad model architecture design or a full-scale data collection.

- Data Constraint

Pre-anonymized publicly available datasets will be the only resource used in the project. Privacy and security laws make it impossible to derive real-world proprietary financial data of institutions available, a situation that can compromise the authenticity of some fraud patterns.

- Computational Resources:

The model expansion and education will take place on the commonly available personal computing hardware and free clouds. This project will not depend upon special high-performance computing clusters.

- Project Goal

It will consist of a report and an academic proof of concept as the key deliverables. The engine developed is not supposed to be deployed on the production level or to be used commercially.

3) What will and will not be done as part of the project

- In Scope
    1) Implement effective fraud detection model, based on both the conventional machine learning and deep learning capabilities.

2) Thoroughly test the models based on the performance on an imbalanced data.
3) Conduct the data pre-processing and feature engineering of the chosen dataset.
4) Design a simple, pictorial user-interface or dashboard that can be shown.
5) Write a well-documented project report.

- Out of Scope

1) The project does not include an implementation of the engine to a live operating system or linking it into the real-time transaction processing pipeline of the financial institutions.
2) No patient-identifiable or sensitive fiscal information will be utilised in the project.
3) The project will neither be tested in real conditions nor tested by A/B.
4) The optional user interface will not be an enterprise level application. This is only used in demonstration purposes.
5) It is not the project that will implement or verify engines on a particular regulatory framework, like GDPR or CCPA.
6) Non-transactional information sources which include social media sentiment, behavioural biometrics and external news events to perform fraud detection will not be explored in the project. It will be kept on the transactional characteristics of data.

1.5 Potential Benefit

- Tangible Benefits

1) The use of behavioural biometric data such as typing or usage of the mouse can be an effective way of enhancing a model that is supposed to identify the account takeover (ATO) fraud. In case the normalization of the speed of typing of a user or his/her route changes considerably, it is a good evidence of fraud.
2) The model will be able to make more intelligent decisions because of contextual data. One example is a user that is traveling in another country posted on their social media that they are on vacation hence a country-country transaction would be less chance of false positives. This minimises the number of genuine

transactions wrongly detected as fraudulent, which saves time by the fraud analysts and enhances the customer experience.

3) Signals that allow prediction can be found by the analysis of exterior news events or social media sentiment. As another example, an increase in social media complaints regarding a certain bank regarding the so-called phishing scams may be used to temporarily enhance security of customers affected by a bank by anticipating a high amount of fraud.

- Intangible Benefits

1) An improving system with a low level of transaction rejection of the legitimate customers is more accurate and proactive. That creates a trust and loyalty since customers will believe that their banks are sufficiently and effectively securing them without unnecessarily putting checks on them.

2) A leading position in fraud prevention can significantly enhance a company's brand reputation. It positions financial institutions as secure and reliable partners in an increasingly risky digital world.

3) The primary role in preventing fraud could substantially improve the brand reputation of the company. It puts financial institutions in a safer, more secure position to be partners in a digital world that is getting riskier and riskier.

- Target Users

1) The biggest users will be banks and credit card firms. They will have the financial advantage of cutting loss of frauds and the non-financial advantage of the gaining of increased customer faith and brand significance.

2) Advanced engine will serve as a selling point with regard to a startup company. It will allow them to provide the newest available security services right upon their establishment, which facilitated building a customer base and reputation on the market.

1.6 Overview of IR

The report submitted in the investigation is the guiding document to the project and it describes, justifies and provides the rough design of the intelligent fraud detection engine. It has a number of main chapters to give the general picture of knowledge and technology basis the project is based on. The report will start with an introduction highlighting the increasing menace of cybercrime in finance and how defence mechanisms have become not enough. It gives the background, statement of the problem, project objectives and goals of the project ensuring the research has a proper purpose. This is followed by the literature review as the main research chapter that should go through several elements of financial fraud in an organized way. Well, this section gives a comprehensive description of various kinds of malware attacks, existing intrusion detection systems based on machine learning, and reasons to switch to deep neural networks and presents its merits. It also examines similar systems and research that relate to each other so as to establish gaps in the literature. The chapter on methodology subsequently describes the plan of the project, which includes the generalization of a hybrid approach of agile and CRISP-DM methodologies. It outlines the major stages of the project, including data understanding, model training and states why this method is perfectly suited to data-driven projects. Lastly, the conclusion summarizes everything documented in the report, affirms the results of the initial study, affirms the logic of the project, and establishes main areas and a gap that could be explored in future.

1.7 Project Plan

Project Plan					
No.	Tasks	Duration (days)	Start Date	End Date	Status
Chapter 1:	Introduction		7/23/2025	8/12/2025	
1	Introduction	20	7/23/2025	8/12/2025	Completed
2	Problem Background	1	7/23/2025	7/24/2025	Completed
3	Project Aim	1	7/23/2025	7/24/2025	Completed
4	Objectives	1	7/23/2025	7/24/2025	Completed
5	Scope	3	7/23/2025	7/26/2025	Completed
6	Overview of the IR	20	7/23/2025	8/12/2025	Completed
Chapter 2:	Literature Review		7/26/2025	8/5/2025	
7	Domain Research	7	7/26/2025	8/2/2025	Completed
8	Similar Systems/Work	2	8/2/2025	8/4/2025	Completed
9	Technical Research	1	8/4/2025	8/5/2025	Completed
Chapter 3:	Methodology		8/2/2025	8/11/2025	
10	System Development Methodology	2	8/5/2025	8/7/2025	Completed
11	Data Gathering Design	1	8/2/2025	8/3/2025	Completed
12	Conduct Survey	7	8/3/2025	8/10/2025	Completed
13	Analysis	1	8/10/2025	8/11/2025	Completed
Conclusion					
14	Conclusion	1	8/11/2025	8/12/2025	Completed
15	Report Formatting	1	8/12/2025	8/13/2025	Completed
16	IR Checking & Submission	0	8/13/2025	8/13/2025	Completed

Figure 1Project Plan

2. Literature Review

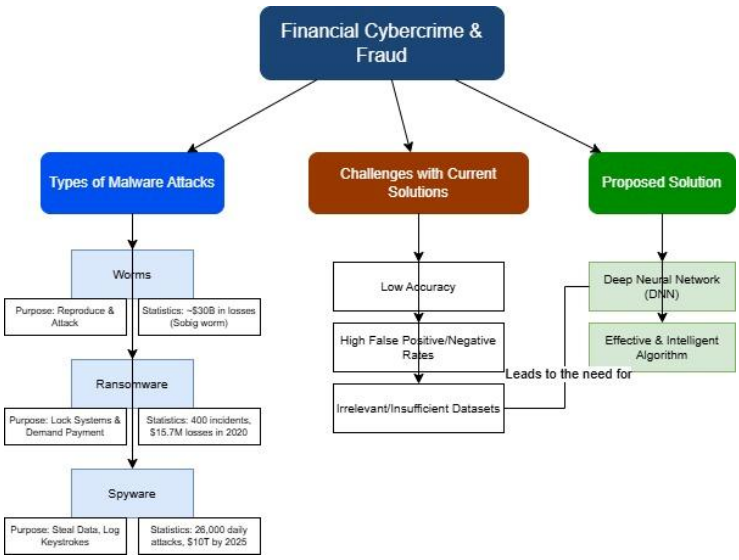

Figure 2 Literature Review

No other type of crime is increasing as fast as that of malware attacks. A report released by Evolve states that financial institutions have been the target of cybercriminals. Malware attacks cost more than 19 percent of financial institutions up to 18.3 billion dollars by 2017. Energy and utilities constitute the second industry that has been mostly impacted. In financial technology, cybercrime can be explained as a form of fraudulent activities executed through the usage of mobile

devices and other forms of technology networking, where they direct their efforts towards disrupting the technology, breaking the reputation of the customers, economic losses, leakage of sensitive information, and resulting in feelings of anxiety. Cybersecurity statistics indicate that losses to cybercrime will hit a high of $23.82 trillion by 2027 and on average, there are 2,200 airline incidents annually that are caused by cyber-attacks, every 39 seconds. Malware is used by cybercriminals to disrupt and destroy servers, mobile devices, and computer networks; encrypt, delete, alter, monitor, and steal sensitive data and registration files; and gain unauthorised access to fintech networks and applications. The malware is in a continuous state of advancement; therefore, it is capable of gaining access into the systems through fooling the authorized users to open infected files, visit risky websites or unintentionally use downloading and installing files via connecting to compromised devices in which to install malware (Ali et al., 2024).

However, the accuracy of the current solutions can be so low with big false positive/negative rates. This is due to the fact that the synthesized attack dataset can be irrelevant and insufficient with extraneous information which makes the machine learning based intrusion detection system (IDS) useless to detect a variety of threats. Thus, an intelligent and effective algorithm like a deep neural network must be developed to face and deal with different sorts of cyberattacks and its variations (Thakkar & Lohiya, 2023).

2.1 Type of Malware Attacks

i) Worms

Computer worms primarily have the purpose of reproducing and attacking other computers, as well as keep running on the computer that it has already compromised. In order to distribute computer worms, the first time, hackers so far send malicious attachments in phishing email or instant messages. Fraudsters tried to conceal the worm to fool victims into launching the software. They accomplished this by using data names and/or double file extensions that seemed important or innocuous, such "invoice." In August 2003, millions of Microsoft Windows computers across the globe were infected by the Sobig computer worm, which resulted in losses of $30 billion in Canada, the United Kingdom, the United States, continental Europe, and Asia. Sobig.A was first identified in January 2003. Sobig.B,

which went out on May 18, 2003 was previously referred to as Paly. The Sobig.C mentioned below published on May 31, 2003, addressed the problem of time with Sobig.B. Sobig.D and Sobig.F would be found a few weeks later. Sobig.F was known to be the fastest-spreading email worm on June 25, 2003. Sobig.F was the most popular and detrimental one (Yan Yan et al., 2022).

ii) Ransomware

Ransomware and malware hit their important destructive form at the financial operations stage sending user data to the sinister symphony. Based on the statistics of the Cybersecurity and Infrastructure Security Agency (CISA), the number of ransomware attacks on financial institutions continues to grow and has hit both the established banks and fast-developing fintech corporations. In 2018, hackers executed 250 ransomware attacks resulting in losses of $9.6 million. In 2019, 300 ransomware attacks resulted in $11.2 million in losses. Ransomware has resulted in a loss of 15.7 million in 400 incidents in 2020. The level of ransomware updates can be evidenced by the following statistics: In 2020, ransomware victims amounted to 400 people, with the losses amounting to 15.7 million. By comparison, 2019 was inconsequential as 300 incidents resulted in $11.2 million economic loss. The rising levels of financial resonance point to the fact that the depth and severity of ransomware attacks are becoming more difficult. (Kamuangu.P, 2024)

iii) Spyware

Spyware is a malicious application meant to track the devices used, normally smartphones. It is a sort of a ghost in your machine. Unauthorized ghosts can also turn out to be harmful to smartphones. Such risks involve accessing the speakers, the microphones of the victim, cameras, keystroke logging, pattern logging, bank credential stealing and crypto mining through mobile phones. Most of the cyberattacks that happen daily about 26,000 are as a result of spyware. The Economic motives of the reasons that drive these intrusions are predominant. Considering the fact that the overall damages are estimated to reach the

astronomical level of $10 trillion by 2025, it will increase exponentially. In one of the surveys, 85 percent of the mobile phones get malware. (Naser, M et al., 2023).

2.2 Machine Learning-Based Intrusion Detection Systems

i) Detecting Low-Rate Distributed Denial of Service (LR-DDoS) attacks

Ransomware, phishing, malware, and distributed denial-of-service (DDoS) are the most widespread cyber threats to financial systems. Also, as a result of the interconnected character of FinTech ecosystem, third-party service providers represent a gate to the network intrusion, therefore, the risk of attack through a supply chain increases (Uma et al., 2024). Phishing, data theft and distributed denial-of-service (DDoS) attacks are common cybersecurity threats within the financial sector. To prevent DDoS attacks, this solution uses machine learning technology and has developed an intrusion detection system. This technology can improve the accuracy of identifying and thwarting low-rate DDoS attacks. In DDoS attacks, traffic is very high. On the contrary, the low-rate DDoS attacks maintain a minimum amount of traffic, which is difficult to be detected.

ii) Anomaly Detection

Anomaly detection is a valuable part of protecting against fraud since it consists of detecting behavioural patterns that are unexpected (see data). Unusual involved transactions can be evidence of a fraudulent transaction that is not in line with the normal transactional trend ("International Journal of Network and Communication Research," 2021). Anomaly-based detectors, which can be configured to measure the typical behaviour of a system to be guarded, will send anomaly alerts when the deviation between observed behaviour (at a particular point in time) and "normal" behaviour surpasses some defined value. One can also simulate the system to behave abnormally by artificially injecting the difference between the desired and observed behaviour and provide an approach to an alarm state when the error margin goes below a set tolerance. Anomaly-based intrusion detection systems are able to detect malware that has gone unnoticed and attacks by extensively using the transmitted pieces of information. RF models are employed to ascertain the value of feature importance in

anomaly detection with the application of weighted k-means technology. To counter these shortcomings and shortcuts, machine learning practices are employed to emulate normal behaviour and establish biases through isolation forests which are a subgroup of support vector machines and local outlier factors.

iii) Signature-Based System

Abuse detection (signature based) is classified into two phases online and off-line. Online stage is designated with detection of misuse based on matching of network traffic against known intrusion patterns (signatures). Otherwise, not corresponding to the above discovered forms of intrusion, it will be defined as novel kind of attack. Intrusion detection systems that are signature-based exploit known attack signatures in order to detect the attacks (Dina et al., 2021). Using these IDs, zero-day attacks or novel kinds of attacks cannot be identified. The anomaly-based intrusion detection employs the unsupervised learning and the signature-based learning normally employing the supervised learning. The easiest form of classification is that of binary classification where two classification labels are employed. Intrusion detection is binary where "normal" and 'attack' (or "abnormal," "anomaly," etc.) are used in binary classification. The creators of malware are continuously coming up with new tricks on how to evade being identified. Malware also has methods called encryption, polymorphism, and obfuscation, which alter the malware and with this make it more difficult to detect through the use of signatures. Polymorphic malware and obfuscation cast doubts on the efficiency of signature-based detection, because polymorphic malware may alter its own code on each infection and obfuscation obscures malicious code by placing it inside legitimate-seeming data (Kothamali, 2022).

2.3 Deep Neural Network

i) Identify malicious activities

Investigation of most of these cyber security problems, including intrusion detection, malware or botnet identification, phishing, network attack prediction (e.g. DoS), fraud, or network anomalies can be accommodated by adopting deep neural

network learning technology, or mainstream use of integrated and hybrid methods. Deep learning can be employed to develop security models since it is more precise, particularly with respect to learning big datasets, which can be security datasets (Sarker, 2021). Now deep neural networks can be simulated with a progressively deeper hidden layer and an increment of neurons on each layer. With such advances, it is now possible to apply deep learning in business settings. Recently there has been evidence of this system to be at par with human experts. Prediction has been attempted with many deep learning architectures, including convolutional structures and neural networks with long short-term memory (LSTM) (CNN). The problem of missing values is highly problematic to solve with the help of deep learning techniques. CNN and LSTM would not allow due to the presence of a lot of blanks in our data. (Mirza et al., 2023).

ii) Prevent cyberattacks

The term deep denotes a sub-discipline of machine learning that applies multi-layer constructs of neural networks to model sophisticated patterns of large data. Such models do not involve direct programmatic instructions to perform some tasks, but they rather learn on their own as they accumulate experience and get better in the future. High-throughput trading environments require deep learning models since large trading data is simple to process and analyse using deep learning models. The fraud detection process using deep learning models is much easier than the traditional methods which rely on tedious feature engineering as they can automatically find pertinent features in raw data. Recently, deep learning has succeeded in being used to detect fraud and has posted impressive results in those tasks that high-dimensional data, including image and speech recognition. The Deep learning models provide an active strategy against fraud since it will learn continually and evolve according to emerging fraud trends. The models allow recognising the minor breaches and thus early intervention, which could indicate new or evolving fraud methods (Bello O.A et al., 2022). Artificial intelligence technologies, including machine learning (ML) and deep learning (DL) have entirely revolutionized the methods applied to detect and prevent fraudulent actions. This not only contributes towards increasing the security of the financial transactions but gives the users a reliable and convenient time. Digital transformation of the financial services industry has resulted in significant efficiency and accessibility of financial transactions; this has been brought about by the emergence of financial technology (fintech). With this shift, however, the complexity of financial fraud cases has also expanded,

especially in its numbers, which presents great challenges to the financial institutions and the customers. The financial industry is a sphere that can be enhanced by the innovative capabilities of artificial intelligence (AI) and nowadays is appreciating its value as a fraud-detecting device. The applicability of AI in fraud detection has unmatched potential to enhance accuracy, speed, and forecasting powers of the systems. Stakeholders have been able to analyse hordes of data, learn and detect intricate fraud schemes, and make judgments on possible fraud through powerful machine learning (ML) algorithms and deep learning (DL) models, which has not been possible before. In addition, fraud detection mechanism systems powered by AI are able to adjust to emerging and evolving frauds. AI systems improve their detection models through each transaction, and are capable of constantly keeping up with the evolving strategies of defence and may be updated to a response by manual intervention where rule-based systems often inherited their lagging between offensive and defensive strategy (Shoetan & Familoni, 2024).

2.4 Similar Systems

Recurrent Neural Networks

A restricted neural network utilizing forward and feedback relationships to compute time sequence information or sequential data is known as a recurrent neural network (RNN). At every loop, an output is added to the input with the addition of new input again in the network. The neural network is also possible to remember the past data when processing the next outputs using the feedback connections. An architecture can also be referred to as a recurrent neural network since this form of processing may be said to be cyclic in nature. Input-output mapping networks are an additional form of RNN that are used to classify sequence data and prediction. The idea of RNNs was introduced by Rumelhart et al. in a published letter in Nature 1986 whereby they deployed self-organizing neural networks, explaining a novel learning process. The Hopfield network can be

characterized as a special form of RNN properly and having symmetrical connections in which the connections of a node to another are equal (symmetric) and vice versa. Another landmark in the history of developing RNNs is its (re)discovery. Owing to the fully connected properties of Hopfield networks, the output of a given neuron becomes input of another neuron and the nodes are updated (binary- 0/1). This kind of network is modelled to specifically replay a human memory (Das et al., 2023).

2.5 Technical Research

This project comprises the smart fraud detection engine and its development presupposes the distinct technical stack, hardware and software. The only major development hardware will be a normal personal computer that is best suited with a graphics processing unit (GPU). The GPU can be used because learning deep learning models can be very computationally demanding since the parallel processing power of the GPU can greatly speed up the training process allowing experiments to be run more quickly. It will be complemented with a use of cloud resources when tasks with considerable computational demand have to be done, e.g., Google Colab. On the development environment, Visual studio code (VS code) will be used as a major integrated development environment (IDE) of choice. It has excellent support of Jupyter notebooks and a lot of extensions making it a perfect choice as an interactive workspace to run data analysis, building models and documentation. The project is launched on the basis of the software founded on Miniconda Python distribution, which could be used due to its executability with creating stable, isolated environments and managing complex library dependencies without any manual installation of compilers, thus preventing frequent build errors. Important libraries and frameworks are Pandas and NumPy to manipulate data and Scikit-learn and Imbalanced-learn to create baseline models and deal with classes imbalances, and TensorFlow with Keras to build and train deep learning autoencoders. Lastly, Matplotlib and Seaborn will be applied towards simple data visualizations that are important to the exploratory data analysis and the results and presentation of the project in the final report.

3. Methodology

3.1 A Hybrid Agile and CRISP-DM Approach

The hybrid approach will include data-driven cyclical phases at the heart of the CRISP-DM and the flexibility and iterative nature of Agile development. The technique can be applied so that the project is more targeted at providing high-quality data-based solutions and can be refined and adapted to new information being learned.

- **CRISP-DM**

The de facto industry standard and process model for implementing data mining initiatives is CRISP-DM. CRISP-DM is a data mining process model that is independent of industry. There are six iterative stages in all, starting with business knowledge and ending with deployment (Schröer et al., 2021). A process model serves as a framework for initiatives and enhances stakeholder comprehension and communication. Several tools, methods, and staff are needed for a data mining process, in addition to effective management. The concepts and ideas of other processes, such KDD and SEMMA, are included into CRISP-DM. Because of its strength, it served as the foundation for more contemporary ideas like the Microsoft Team Data Science Process (TDSP) (Shimaoka et al., 2024).

- **Agile**

Agile, which has its roots in the IT industry, places a strong emphasis on adaptability, early delivery, and improvement over multiple cycles in cooperation with project stakeholders. Agile is about a culture shift as much as a shift in reality and business procedures. The FinTech sector needs to break the mould, decentralise decision-making, and embrace a coaching and feedback culture when integrating AI. Agile is increasingly being used by banks, FinTech, and insurers who want to create more customer-focused solutions, shorten feedback loops, and speed up innovation. Additionally, Agile allows for the use of cutting-edge technologies like cloud computing, data analytics, and artificial

intelligence—all of which are now essential to contemporary FinTech operations. Following that, frameworks that incorporate various practices and structures according to certain organisational contexts are used to implement agile approaches. Scrum is among the most well-known agile frameworks as a result. It has distinct roles, such as Product Owner, Scrum Master, and Development Team, and operates in brief development cycles, or sprints, that run two to four weeks. The Scrum rituals of sprint planning, daily stand-up, and sprint review promote accountability, openness, and ongoing development (Goyal, 2025).

Phase 1: Business and Data Understanding (CRISP-DM)

The aim of this project is to conceive and create a smart fraud detection engine that employs deep learning technology so as to efficiently detect and stop financial fraud transactions which will increase the security and integrity of online financial networks.

- Project Objectives

1) To identify and categorize most significant and forthcoming cybersecurity risks in FinTech.
2) To assess and examine the effectiveness of the defensive strategies that financial institutions can employ.
3) To suggest improvements to the current counter-defensive strategies that will enable them to better handle the complexities and shifts of the new cyberthreats.

- Problem Statement

Malware is one of the greatest threats to cyberspace. Even better, as Evolve Taping sets forth, malware most often preyed on the Financial Services sector (where they had represented 19 percent of all malware attacks and generated more than $18.3 billion in the losses incurred during the year 2017 alone). The worst was experienced in the utilities and the energy sectors. Using the FinTech ecology context, cybercrime herein can be defined as any fraudulent activity conducted over

a mobile phone and other networking technologies to either destroy the equipment, embarrass the customers, cause monetary loss, steal confidential information, and cause fear and panic. According to cybersecurity research, cybercrime will cost $23.82 trillion by 2027, with 2,200 cyberattacks on average every 39 seconds. Hackers can use malware to compromise and take down servers, mobile devices, and computer networks; encrypt, delete, change, spy on, and steal registration files and private information; and even get unauthorised access to finance networks and apps. Malware is dynamic and can infiltrate a system by deceiving authorised users into opening dangerous files, by accessing systems where malware has been installed without authorisation, or by downloading and installing malware on machines that have been compromised (Ali et al., 2024). There is the possibility of using machine learning based intrusion detection systems (IDS), especially to detect Low-Rate Distributed Denial of Service (LR-DDoS) attacks. To enable the accurate detection of these attacks at a high rate, these systems utilise the application of tree based classifiers to do so including Random Forest (RF), Support Vector Machine (SVM) (Javaheri et al., 2023).

However, there is a chance that this present approach will have a low false-positive or false-negative ratio. This is due to the possibility that attack datasets contain a large amount of redundant and superfluous data, rendering ML-based intrusion detection systems (IDS) ineffective at identifying a wide range of threats. Thus, the development of a clever and effective mechanism such as a deep neural network is one way to combat different types of network attacks and their variations (Thakkar & Lohiya, 2023).

- Data Source:

The dataset employed in this project was extracted by one of the most known public database repositories, Kaggle. This decision was done to accommodate project constraints especially because of the strict privacy laws and security concerns which did not allow access to proprietary and actual financial information. As the project aims at the development of a fraud detection engine, the set of anonymous transaction data is quite suitable to the selection. This method will help

to ground the research in an effective and reasonable source of information and at the same time satisfy any confidentiality and ethical consideration.

Phase 2: Data Preparation (CRISP-DM)

- Exploratory Data Analysis:

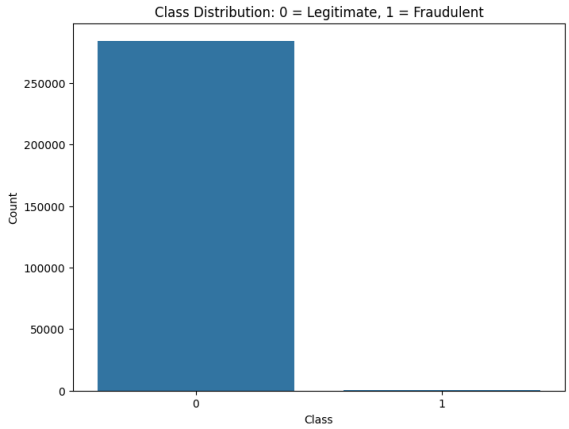

Figure 3 EDA

A significant observation using exploratory data analysis (EDA) of transactional records is the overwhelming nature of the class imbalance on the transactions between genuine and fraudulent ones. This skew is also characteristic of the real world data of fraud in the sense that telecommunication frauds happen rarely as compared to the high numbers of known legitimate transactions.

In order to measure this the dataset was examined to ascertain the precise proportion of every category. The findings (quantities) indicated that class 0 (legitimate) of all transactions was overwhelming and class 1 (fraudulent) was incredibly on the bottom. This is visualised in the bar chart below.

The apparent distinct difference between these two classes as listed in the above figure presents a significant problem to the supervised machine learning models. Unless correctly used, the model may end up with very good accuracy by merely assuming that every transaction is valid, which is indeed true over 99 percent of the data. Nonetheless, this model will not provide any visibility to the fraud, which will be meaningless to the project goals. Consequently, the correction of this

imbalance will play a key role in the data preparation step since it will help in the effectiveness of the model.

- Data Cleaning and Preprocessing:

```
df.drop_duplicates(inplace=True)
```

Figure 4 Data Cleaning and Preprocessing

The dataset is cleaned first before carrying out any transformations. The initial process would be to find and clear the duplicates so as to have all the records to reflect unique financial transactions. This is done with the help of drop_duplicates() procedure in pandas library which is important to retain the integrity of the data and avoid model bias.

- Feature Engineering:

```
# Extract features like hour of the day
df['Hour'] = df['Time'].apply(lambda x: x // 3600 % 24)
# You could also add features like day of the week if the data spans multiple days
```

✓ 0.0s Python

Figure 5 Feature Engineering

After the first stage involving the cleaning and preprocessing of data, raw features designed based on the dataset were created to improve the model on identifying the fraud pattern. Such process is essential since it permits the model to learn based on a more meaningful representation of the data than in the original form. Columns V (V1-V28) are anonymous in the dataset which implies that conventional feature engineering cannot be applied extensively, as it is the focus on available time and quantity features.

Time feature is the number of seconds that have passage since the initial transaction of the set of data. It does not come in an immediate form. Rather than this, a new value, Hour, is applied to signify the of the day. This is computed by multiplication of the total time (in seconds) with seconds in a day. The concept

behind it is the time-based patterns of fraud since they are more probable to take place within certain non-work hours.

This has not been applied to the code supplied but you may want to consider as a possible next step or in your report. You may speak about how such features as the ratio between the frequency of transactions or the amount of transactions in the short term (e.g., the last 24 hours) and the usual average value of defraud occurred can be a powerful indicator of fraud.

- Handling Imbalanced:

```python
from imblearn.over_sampling import SMOTE
from collections import Counter

# Display original class distribution
print('Original dataset shape %s' % Counter(y))

# Apply SMOTE to the dataset
sm = SMOTE(random_state=42)
X_res, y_res = sm.fit_resample(X, y)

# Display resampled class distribution
print('Resampled dataset shape %s' % Counter(y_res))
```

[9] ✓ 7.4s Python

Figure 6 Handling Imbalance

This was the greatest obstacle that was registered in the data analysis phase as there was severe class imbalance in the collected dataset. Fraudulent transactions (class 1) are so exceptional, as illustrated in Figure 1, such that they form a very small part of the total transactions. This skew is a problem to traditional machine learning algorithms that often tend to classify new data as belonging to the majority class (legitimate transactions) and can therefore erroneously classify almost all of the transactions to the legitimate category and miss the fraudulent behaviour that it is supposed to detect.

As a solution to such a problem, Synthetic Minority Oversampling Technique (SMOTE) was used to create oversampling. SMOTE is implemented by producing synthetic data of the minority groups instead of merely duplicating the available data. It does this by considering the existing points of minority, and then extrapolates synthetic points along lines between the points. The process performs

well to balance the data as the model has access to a more equal distribution of legitimate and fraudulent transactions to learn.

The Imbalanced Learning Python library is implemented as SMOTE. The training data is processed using the SMOTE operation to give a resampled training set where the fraudulent transactional data has an almost comparable ratio with the valid transactional data. This procedure can give a strong basis over the model training stage to make sure that machine learning and deep learning models will not be aiming toward the most dominant category and are more capable of learning the niche aspects of the fraudulent activity.

Phase 3: Iterative Model Development (Agile Sprints)

The steps involved in this part of the project are the development of and training of key fraud detection models. In order to have a strong baseline, conventional machine learning algorithm and advanced ensemble techniques had been applied and tested before deploying deep learning architectures. There was oversampling of the training dataset (X) which trained all models Still, in order to limit extreme class imbalance, train_res, y train_res was performed, and its performance was estimated on the original imbalanced test set, X test, y test so that the actual effectiveness of the method could be measured realistically.

- **Sprint 1: Baseline Model**

1) Logistic Regression:

```
Logistic Regression Results:
Confusion Matrix:
[[82624  2352]
 [   17   125]]

Classification Report:
              precision    recall  f1-score   support

           0       1.00      0.97      0.99     84976
           1       0.05      0.88      0.10       142

    accuracy                           0.97     85118
   macro avg       0.53      0.93      0.54     85118
weighted avg       1.00      0.97      0.98     85118
```

Figure 7 Logistic Regression

The first benchmark that would be used is a linear model to be used to set a benchmark performance score. This model also gives us a straight forward, interpretable starting point with which more complicated models to compare (Salama, 2024).

The model correctly detected 125 fraudulent out of 142 transactions which gives a recall of 0.88. Nevertheless, its accuracy was very low with a value of 0.05 signifying that 5 of every 100 transactions estimated to be fraudulent, were in the real sense fraudulent. The recall rate in this model was good but the false positive rate rendered its application in real life highly impractical.

2) Random Forest Classifier:

```
Random Forest Results:
Confusion Matrix:
[[84965    11]
 [   35   107]]

Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00     84976
           1       0.91      0.75      0.82       142

    accuracy                           1.00     85118
   macro avg       0.95      0.88      0.91     85118
weighted avg       1.00      1.00      1.00     85118
```

Figure 8 Random Forest Classifier

This robust type of integration is an add-on to the linear models. Random forests emulate several different decision trees and then, average the outcomes, which aids in overfitting and better captures the nonlinear tendencies in the data as well (Adugna et al., 2022).

This mode was more accurate with a rate of 0.91 which meant that it is more accurate in detecting fraud and few genuine transactions were deterred in a false positive. It had a recall rate of 0.75, implying that it correctly classifies 107 fraudulent transactions on the total of 142 fraud transactions. It is indicated in the high F1 score of 0.82 which shows the correct balance between precision and recall.

3) XGBoost Classifier:

```
XGBoost Results:
Confusion Matrix:
[[84950    26]
 [   31   111]]

Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00     84976
           1       0.81      0.78      0.80       142

    accuracy                           1.00     85118
   macro avg       0.90      0.89      0.90     85118
weighted avg       1.00      1.00      1.00     85118
```

Figure 9 XGBoost Classifier

XGBoost is the state-of-the-art ensemble model which is the highly optimized gradient boosting framework. XGBoost is known to be fast and delivering superb results on structured data, and will likely come to be a powerful baseline against which deep learning models will be measured (Dhaliwal et al., 2018).

The model precision is 0.81, the recall 0.78, the F1 0.80. Its recall rate is slightly higher even though its accuracy level is marginally lower than that of the random forest. The model accurately classified 111 of 142 fraudulent transactions as such which is a solid representation to indicate that the model is performing well.

- **Sprint 2: Deep Learning Model**

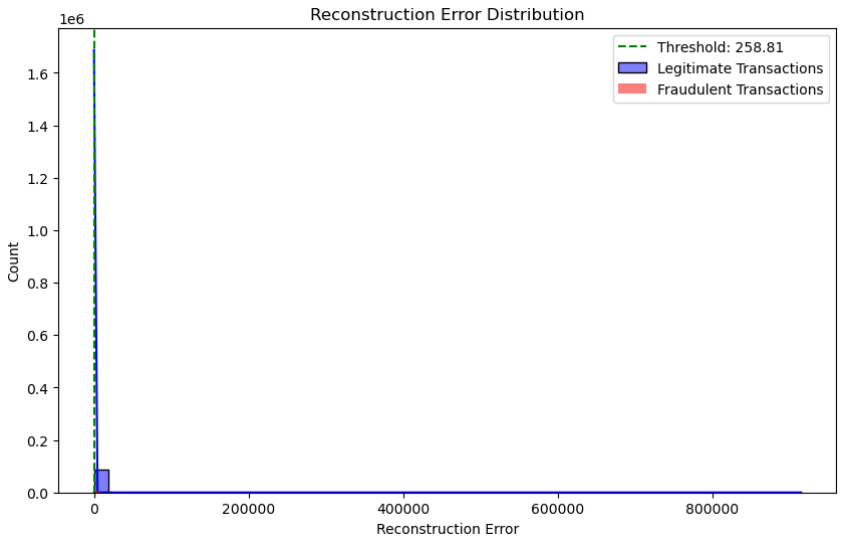

Figure 10 Deep Learning Model

In order to test deep learning, we trained an autoencoder to identify abnormal points in the data. Model was trained only on legitimate transaction (Class 0) hence was capable of learning how to compress and reconstruct normal pattern of transactions.

The above figure, figure 6, reconstruction error distribution, viscerally explains or demonstrates the roving principle behind this technique. It graphs the actual values of the reconstruction errors of legal transactions and fraudulent transactions. The blue distribution is the legitimate transactions whereby the reconstruction error rises frequently recording low reconstruction errors. This means that the autoencoder has learnt to reproduce normal transactions it has been trained on.

The red plot, which indicates unduly transactions, indicates that more high reconstruction errors are distributed. The reason is that, fraudulent transactions are abnormal patterns which will not be discovered by the model since it has never been encountered.

The limit beyond which 95 percent of irregularities are reconstructed is 258.81 (mean of legitimate errors). Any transactions which have an error above this limit are identified as fraudulent.

Phase 4: Evaluation and Reporting (CRISP-DM)

This project phase aims at an overall interpretation of the outcomes of the different models that have been formulated. The primary goal is to rate and compare the performance of all models so as to come up with the best solution to the fraud detection engine. The last step during this stage is synthesis of all the findings, which are now incorporated into a report to present the final project.

- **Comprehensive Model Comparison**

All the models were compared directly using key performance measures used in imbalanced datasets (logistic regression, random forest, XGBoost, and Autoencoder). The table below is a summary of the output of the original imbalanced test set with a particular attention being placed on the accuracy, recall, and F1 score of the type of fraud (class 1).

Model	Precision (Class 1)	Recall (Class 1)	F1-Score (Class 1)

Logistic Regression	0.5	0.88	0.10
Random Forest	0.91	0.75	0.82
XGBoost	0.81	0.78	0.80
Autoencoder	0.00	0.06	0.00

Table 1 Model Comparison

- Logistic Regression

A very low level of precision implies that it has a high false positive rate and logistic regression is not practically usable since analysts might get lost in false alarms.

- Random Forest and XGBoost

Firstly, these ensemble models give a great tradeoff between precision and recall, with great F1 scores. Random Forest has the best accuracy but XGBoost has a little better balance. Such models present a quality and efficient method.

- Autoencoders

The behaviour of auto encoders with respect to the detection of the anomalies needs to be studied. To understand whether unsupervised anomaly search techniques can offer a superior solution, their findings must be compared with comparative applications that take the form of the Random Forest and XGBoost parameters of benchmark tests.

3.2 Analysis

Qualitative and quantitative data collection methods are used in the system. Qualitative data collection entails building qualitative data like experience of the user, whereas quantitative data collection aims to count the responses of the user.

The survey provides quantitative information to users and non-users (financial institutions) of the DNN-based financial technology services to gauge their needs, inconveniences, and satisfaction towards the services. As an illustration, it samples customer satisfaction with DNN-based financial technology software with regard to fraud detection.

- Question 1:

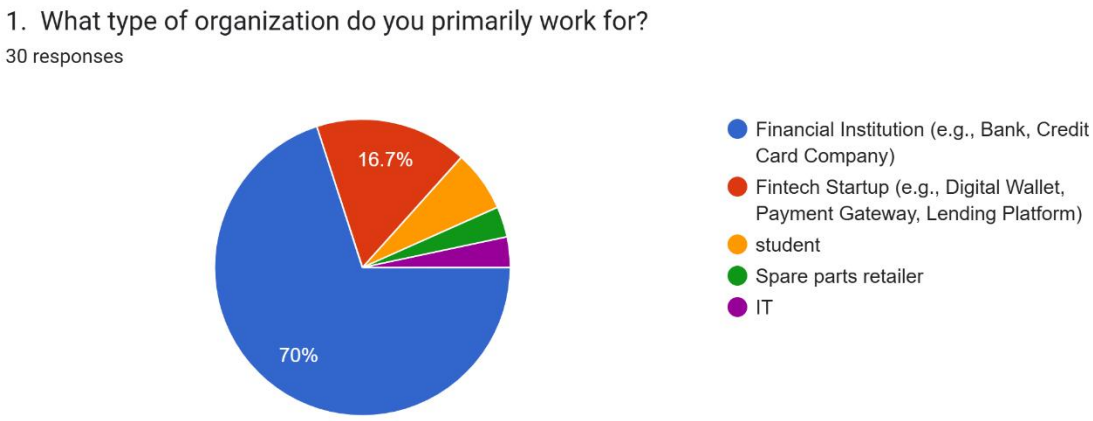

Figure 11 Question

In question 1 that concerned the respondents of the kind of organization where they do most of their work, 70 percent belonged to financial organizations like banks and credit card companies. This made sure that the survey information was greatly ridden with the views of the beneficiaries of fraud detection engines. Also, 16.7 percent of the respondents represented the fintech startups, the acute, quick-developing market branch of the financial sector. Other categories including students, parts retailers and the IT had the remaining responses and they contributed a small percentage of all the responses. This dispersion confirms the efficiency of the survey and makes it clear that it reached the audience that is highly relevant and niche.

- Question 2:

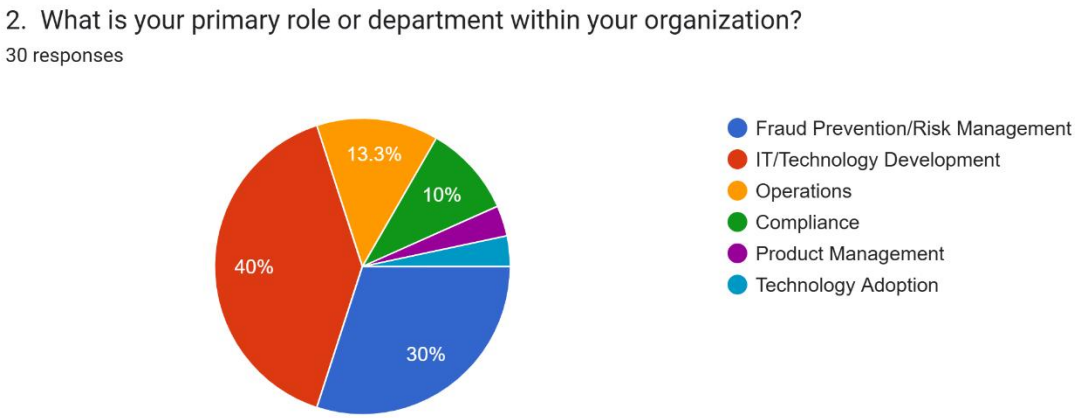

Figure 12 Question 2

Answers to Question 2 were aimed at determining the major roles of the participants and they presented an optimal combination of technical and business understanding. The majority of respondents (40%) were found to be in IT/technical development which helps to offer essential technical information with respect to nature and difficulties of achieving new systems implementation. The next user group of 30 percent of respondents in fraud prevention/risk management was a key target audience whose daily fraud experiences hence have a direct reflection on the project goals. The rest of the answers were assigned to operations (13.3%), compliance (10%) and other departments related to practitioners in making sure the extensive and holistic representation of finance and fintech ecosystem.

- Question 3:

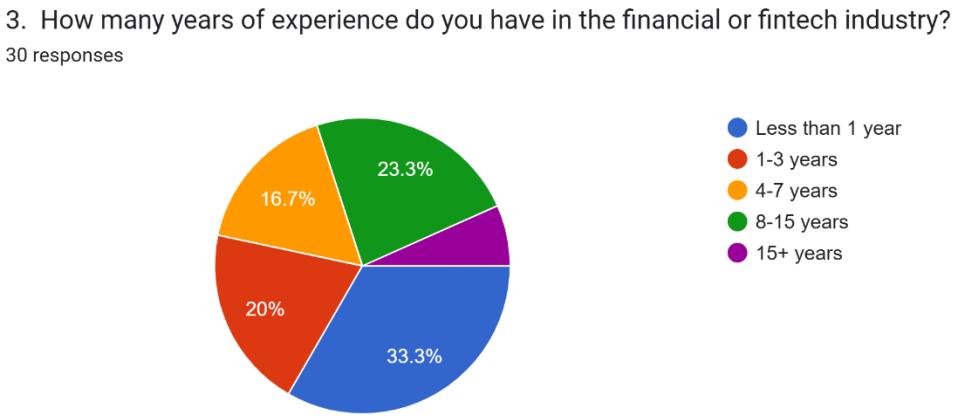

Figure 13 Question 3

In Question 3, 33.3 percent of the participants possessed less than 1 year experience of working in the (financial/fintech) sector. This gives important revelations by people who could be conversant with the current trends in technology. Further, quite a number of mid-career professionals were surveyed including 20% with 1 to 3 year work experience, and 16.7 % with 4 to 7 years work experience. More importantly, the survey would have the views of experienced professionals as well of whom 23.3 have more than 8-15 years experience and 6.7 percent have more than 15 years experience. This wide coverage of both the novice and the experienced makes the survey based on different ways of thinking, new knowledge of technology to long-term and highly experiences in the industry.

- Question 4:

Figure 14 Question 4

Question 4 determined the relevance of financial fraud as a challenge, and the answers to Question 4 confirmed the premise of this project in a strong way. At least 90 percent of respondents were of the view that financial fraud is very or at best seriously wrong (30 percent and 50 percent respectively). The rest of 10 per cent considered it as slightly significant and none has it as not significant or very significant. Such a huge majority demonstrates that the problem of financial fraud is not underestimated anywhere in the industry and prompts the necessity of more progressive and efficient detection systems. These statistics offer a compelling argument to come up with ways of responding to this dilemma.

- Question 5:

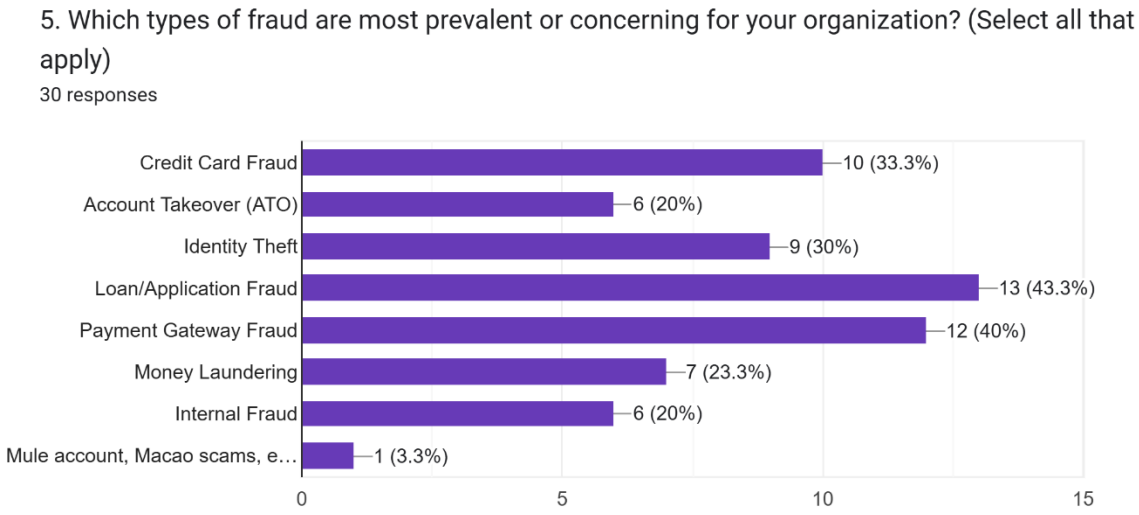

Figure 15 Question 5

Question 5 required writing about the most frequent kinds of fraud, and the discussion of the Question 5 helped to get important information on the particular threats, which should be covered by the new detection engine. The results of the survey revealed that issues of digital financial services have emerged clearly. The most widespread issue is procured to be loan/application fraud as 43.3 percent of respondents chose this issue. Payment gateway fraud with (40%) and credit card fraud 33.3 appeared next. Other threats of great concern are identity theft (30\%) and money laundering (23.3\%). These responses are so frequent, and this justifies

why a good fraud detection engine should be capable of handling a number of complicated digital fraud schemes, especially those that are transaction based and applications based.

- Question 6:

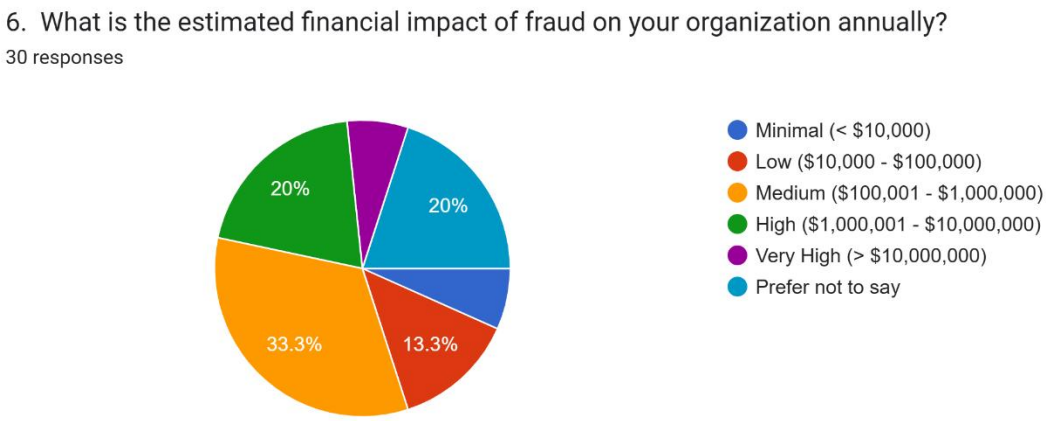

Figure 16 Question 6

The answer to Question 6 pointed to the high financial costs that would be experienced within organizations as reported by 66.6 percent of the respondents whose annual costs of fraud were between the levels of 100,001 and more than 10 million. In particular, 33.3 percent of the respondents noted that the impact was moderate (USD 100,001 to USD 1,000,000), and 20 percent indicated a high impact (USD 1,000,001 to USD 10,000,000). Still another 20% chose the “minimal” (less

than 10,000), and a small percent (13.3) chose the low impact. Importantly, none of the respondents choose the option of very high which is (10 million) and 20 percent chose not to state. These statistics make it clear that the project offers a strong financial justification since a myriad of organizations are suffering massive financial losses because of fraud.

- Question 7:

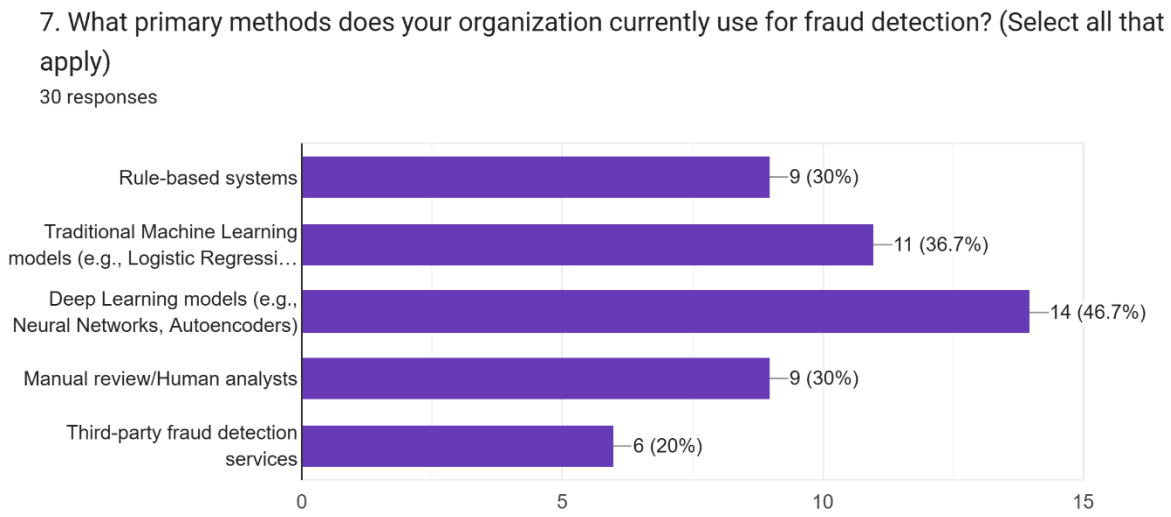

Figure 17 Question 7

Question 7 referred to the main techniques of detecting fraud. Responding to the Question 7, one could identify the unambiguous tendency in the industry toward more advanced technological solutions. The most preferred was the deep learning models where 46.7 percent of respondents said that they apply deep learning models. This fact proves the emphasis of the project powerfully. 36.7 percent of the respondents chose traditional machine learning models as they are also common. Intriguingly, at the same time the implementation of advanced technologies rises, the use of classic approaches becomes also prevailing 30 percent of respondents opted on rule-based systems and manual review/human analysts, meaning that the classic approaches are still in use, with or without the additional use of more sophisticated systems. This implies that creation of a system that will not only perform on exceptional levels but also be easily integrated into the current operating workflows is the context that is worth taking advantage of.

- Question 8:

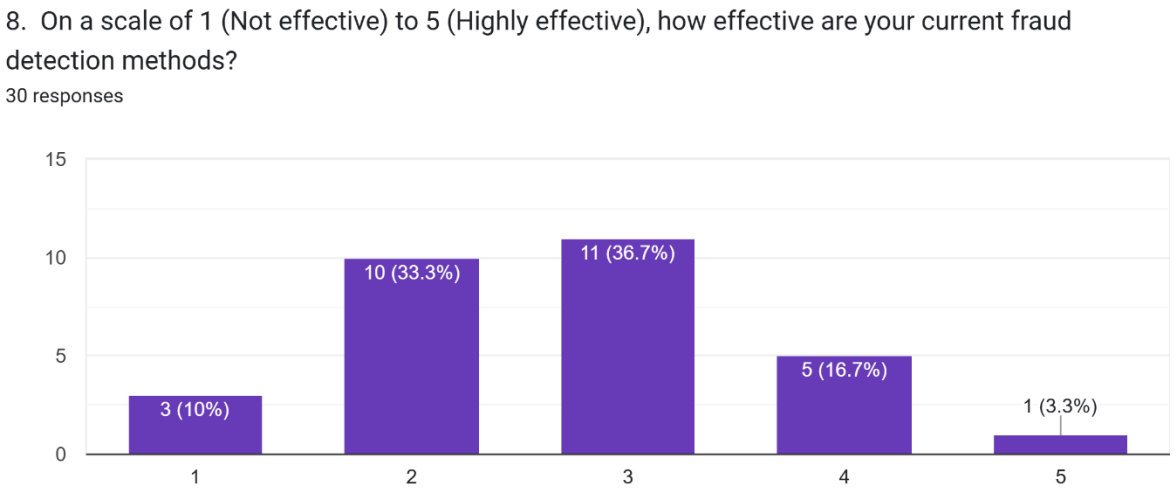

Figure 18 Question 8

Question 8 measured the level of efficiency of the existing fraud detection mechanisms in the scale of 1-5. The indications given in the responses show that there is different opinions on the performance of these methods yet the responses are generally moderate. The biggest percentage of respondents (36.7%) gave their current approaches the rating of 3 (moderate effectiveness). It implies that the current solutions are not totally failing, however, there is a lot that could be done better. The rankings of the respondents revealed that 43.3 percent of the respondents gave a low score of 1 or 2 to their methods, which mean that a large sector of the industry is ineffective. A very few respondents reported their method as being "4" (16.7%) or a "5" (3.3%). These facts support the idea of the need to find a more optimal solution and confirm the insufficient condition of the existing approaches,

which is also evidence of the effectiveness of the project initiative to create a more powerful engine of detecting fraud.

- Question 9:

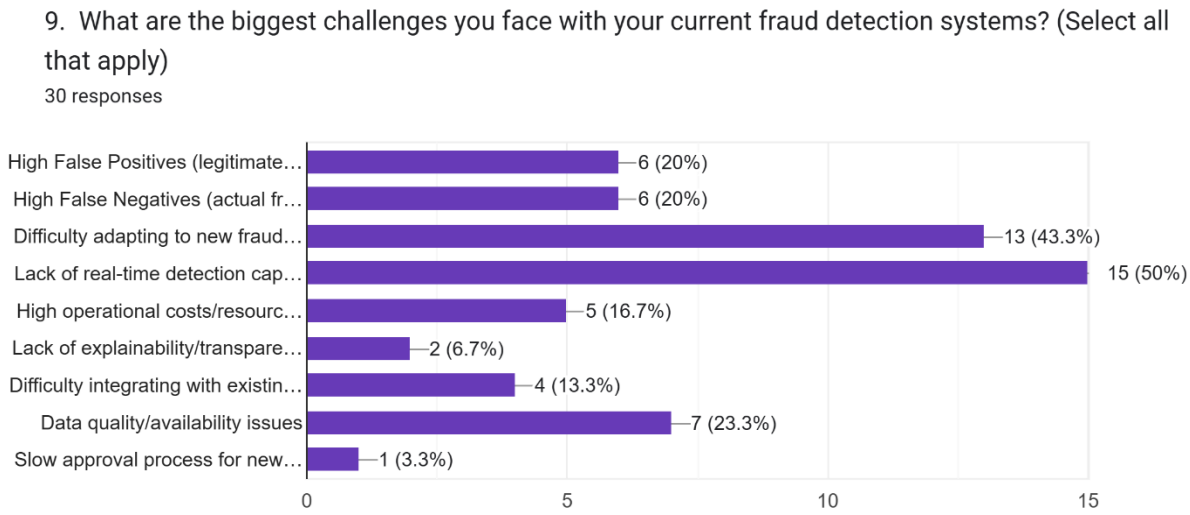

Figure 19 Question 9

Question 9 inquired about what was perceived to be the greatest barriers that were presently confronting fraud detection systems, and answers gave emphasis to the need to be faster and more adjustable. The largest problem was described by the respondents as the issue of not having real-time detection capabilities, 50 percent of the respondents gave this answer. This establishes the fact that the latency of fraud detection is a significant source of pain. Right after is the one related inadaptability of new fraud patterns, where almost half of the responders choose this answer proving to be on the right track where deep learning is concerned to detect almost invisible and changing threats. The rest of the notable difficulties are related to the quality/availability of data (23.3%), the high rates of false positives and false negatives that are both picked by 20 percent of respondents. The results further support the main idea of the project as it is aimed to develop a system that is not merely extremely precise but also futuristic and flexible.

- Question 10:

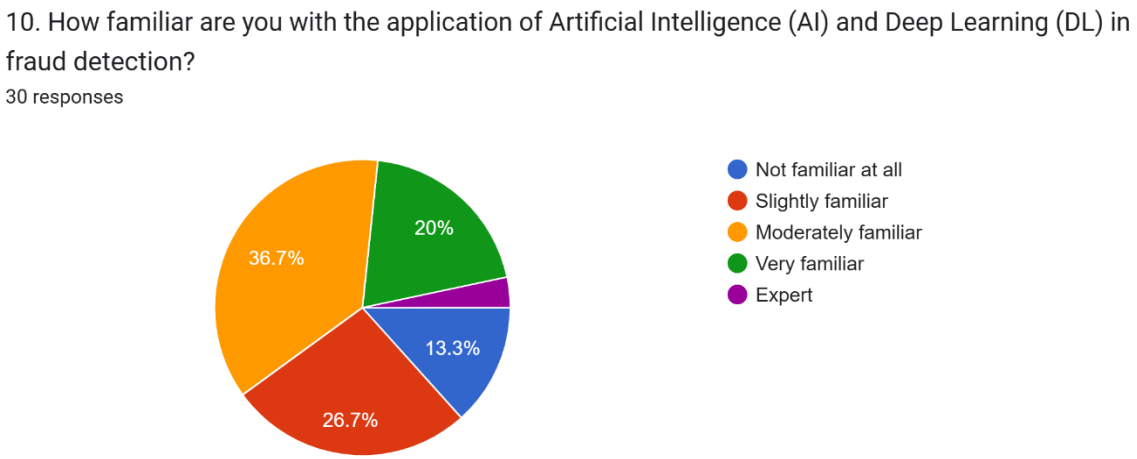

Figure 20 Question 10

The answers to Question 10 determined how well people were aware of the use of artificial intelligence and deep learning in fraud detection, which established that the industry was conversant with the technologies. The 73.4 percent of respondents reported being, first and second, “fairly familiar (36.7)and very

familiar (20). The percent of the respondents to identify themselves as experts was 3.3%. This means that the target population possesses well-developed knowledge base that can be of great help to the projects related to advanced deep learning technologies. This discovery implies that the project does not have to spend time to show how to use artificial intelligence/deep learning since the targeted audience has already accepted and understands its potential.

- Question 11:

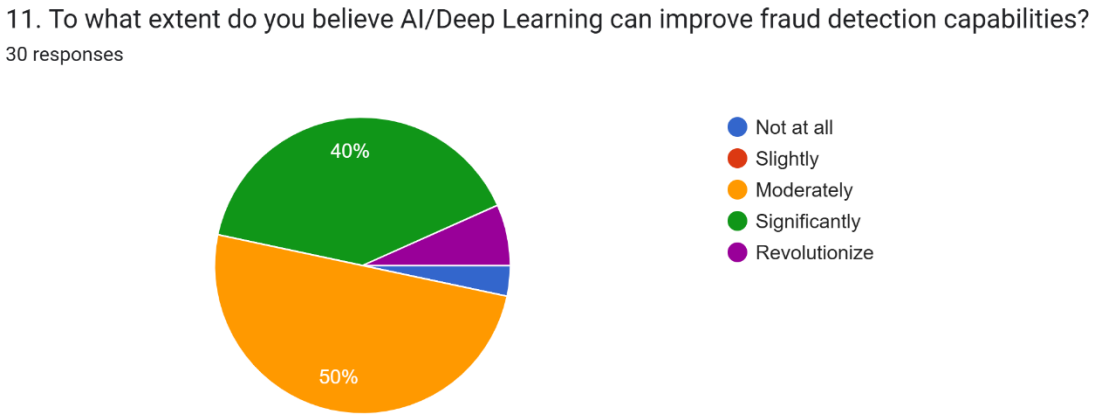

Figure 21 Question 11

Question 11 was the question respondents were asked to what degree they thought artificial intelligence/deep learning could enhance the detection of fraud, and the answers to this question expressed a very strong and positive perspective. Ninety percent of participants harbored the view that the technologies would help enhance the capabilities either moderately (50%) or significantly (40%). A more significant, although small, number of respondents (6.7%) think that AI/deep learning will change the entire sector, which further demonstrates the high levels of expectations toward the technology. The fact that no negative responses received supports the opinion that the industry is willing to accept AI solutions. This observation highly affirms the deep learning technique of the project because the prospective users have already expressed that it can be of great value to their current systems.

- Question 12:

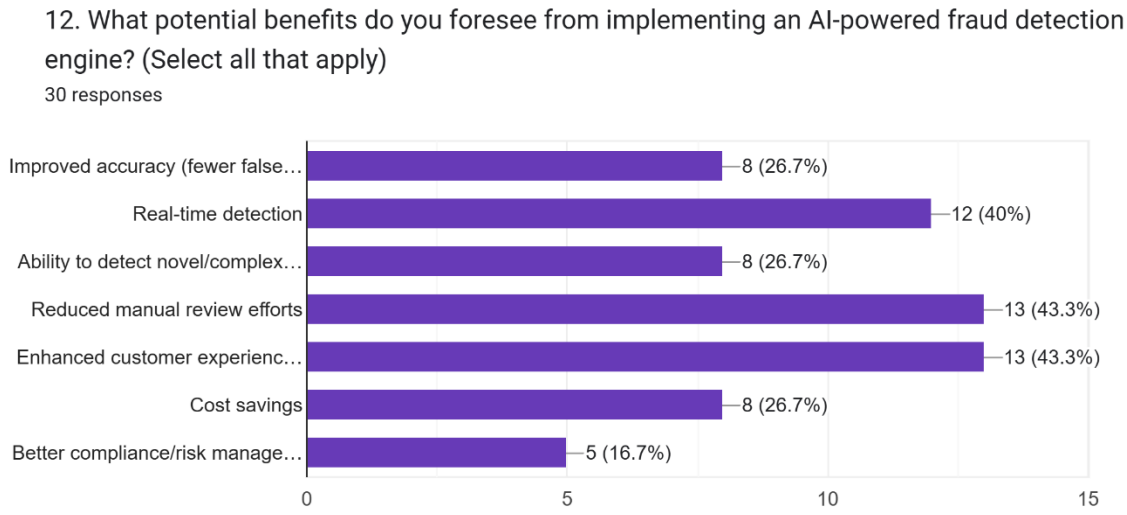

Figure 22 Question 12

Question 12 asked about the potential benefits of an AI-driven fraud detection engine, and the responses to Question 12 revealed a clear set of priorities. The most desired benefits were "reducing manual review work" and "enhancing customer experience," with 43.3% of respondents selecting both of these benefits. This indicates a strong desire for the system to not only improve security but also simplify operations and increase customer satisfaction. "Real-time detection" was also a top priority, with 40% of participants selecting this option, further confirming the conclusions from Question 9 regarding speed requirements. Other significant benefits included "improved accuracy," "the ability to detect novel/complex fraud patterns," and "cost savings," all of which were chosen by 26.7% of respondents. These findings provide a clear roadmap for the project, confirming that a successful solution must possess these key advantages to be adopted by the industry.

- Question 13:

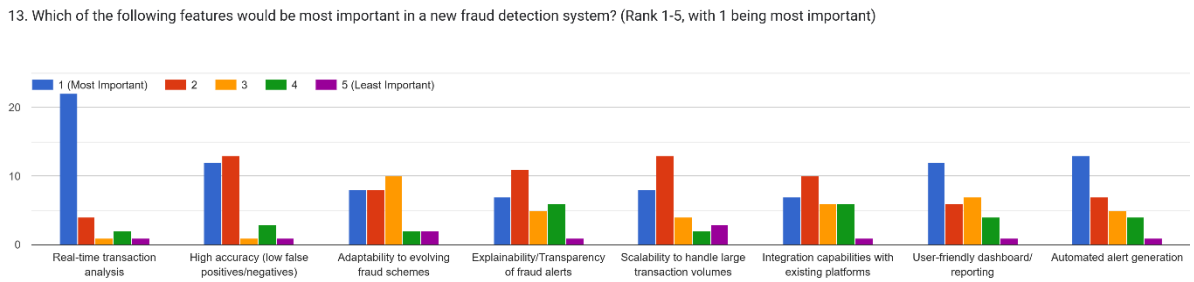

Figure 23 Question 13

Question 13 involved asking the participants to choose the most important features within a new fraud detection system and the answers to Question 13 clearly showed the hierarchy of the priorities. The analysis of transactions in real-time use was ranked as the feature with the major importance, and participants rated it on the maximum level of the setting just as others. This justifies strongly the investor in the project of coming up with a system that is real time. The other features that touched the top grades with a large number of votes in the top two categories are high accuracy, scalability and automatic alert generating, which are in top ranks in their own way. The ease of use of dashboards and flexibility to changing fraud schemes was also considered to be of utmost relevance. On the other hand, explainability/transparency and integration capabilities were only slightly less significant with a high level of attention, which means that even though performance stays the primary concern, usability and adaptability are equally fundamental. Such data will give a significant prototype into the end design and utility of the fraud detection machine.

- Question 14:

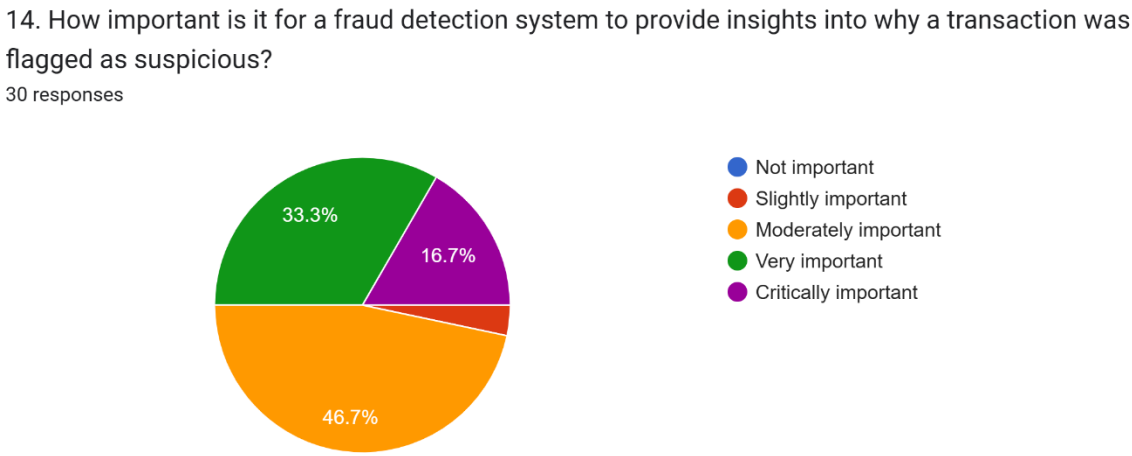

Figure 24 Question 14

Question 14 relied on the significance of model interpretability and the results of Question 14 indicated that there is a robust emphasis on transparency in detection systems of fraud. Eighty-three point four percent of all respondents ranked it as being of moderate importance (46.7 percent), great importance (33.3 percent) or utmost importance (16.7 percent). The above findings are substantial as they show that performance is a priority, but the functionality that enables the system to give insight on why transactions are flagged is an essential determinant in adopting the system in the real world. This input raises a major design issue to the project, the necessity to have a solution that would utilize the weight of deep learning but would have a certain level of understandability that will allow human analysts to take part in the reviewing process.

- Question 15:

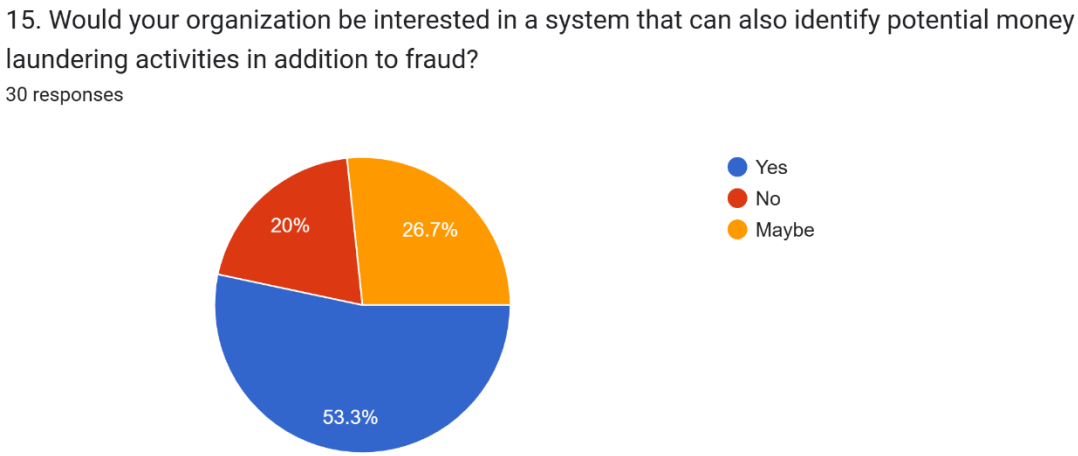

Figure 25 Question 15

The question 15 inquired on whether the industry was interested in a system capable of detecting money laundering. Question 15 was answered clearly where there is an appeal of the existence of a strong explicit need to multifunctional solutions. Most respondents (53.3%) responded with a yes, which signifies that they are quite interested in a system that will combine fraud and money laundering detection. It is interesting to note that 26.7 percent of the respondents said that they are not sure, which means that with time or visualized value they might be convinced to believe in such advantages. Just about one-fifth (20 percent) of those surveyed replied no, which means that a big part of the industry feels that there is a definite linkage between the two kinds of financial crimes. The finding can be illustrated by the assumption that the project under consideration is primarily concerned with fraud; however, other types of financial crimes, including money laundering, can be added to the list of features to implement within the framework of future growth as well.

Chapter 4

CHAPTER 4: DESIGN AND IMPLEMENTATION

4.1 Introduction

This chapter gives detailed reports of the technical architecture, design approaches, and implementation plans used in the development of the intelligent fraud detection engine. The main aim of this stage is to bring the project out of board research that was developed in the previous chapters and to actualize the project into a fully operation prototype. In the chapter, the system is broken down into its main building blocks: a modular system architecture that enables real-time communication, non-relational data schema database that is capable of high speeds of data ingestion and a hybrid intelligence layer that fulfils the security protocols. It also describes how the main features are installed such as a Web Application Firewall (WAF), Multi-Factor Authentication (MFA), and how machine learning inference is incorporated into a Python-based application server.

4.2 Design

The system is built on a more contemporary, decoupled client server architecture to make the user interface responsive and leave computationally intensive processes to specialized backends.

System Architecture: The architecture is further sub-divided into three layers: presentation layer, application logic layer and a data persistence layer.

- The presentation layer is the interface with the client and the interface is created using standard HTML5 and JavaScript. It uses the Tailwind CSS toolkit (Tailwind Labs, n.d.) to make sure that the design is responsive to different screen sizes to provide a user with an excellent experience both when using the desktop and mobile banking.
- A Flask (Python) server is used to drive the Application Logic Layer (Pallets Projects, n.d.). This simple model has been selected because it is compatible with data science libraries (pandas, numpy, scikit-learn), which allows easily incorporating machine learning models into the HTTP call cycle. This layer will

act as a centralized API gateway, which will process all the transactional requests (/predict) and implement security policy prior to any data being stored.

- The data persistence layer leverages on Google firebase, which is the Cloud firestore NoSQL database (Google, n.d.). The reason behind this decision was the necessity in real time data synchronization, event-driven listeners in Firestore allow dashboards to reflect immediately when the status of transactions changes and avoid making manual page refreshes.

Security Architecture: A sound security design policy was embraced. The application also incorporates the custom Web Application Firewall (WAF) middleware at the API entry points. The component is the initial defense mechanism because it verifies incoming JSON payloads and query parameters with a dictionary of known threat patterns, including SQL injection and cross-site scripting. Also, the security of individual sessions is implemented with the help of a script of idle detection, which automatically ends the user session after 60 seconds of inactivity, which will reduce the possibility of unauthorized access to the equipment left unattended.

4.2.1 Use Case Diagram

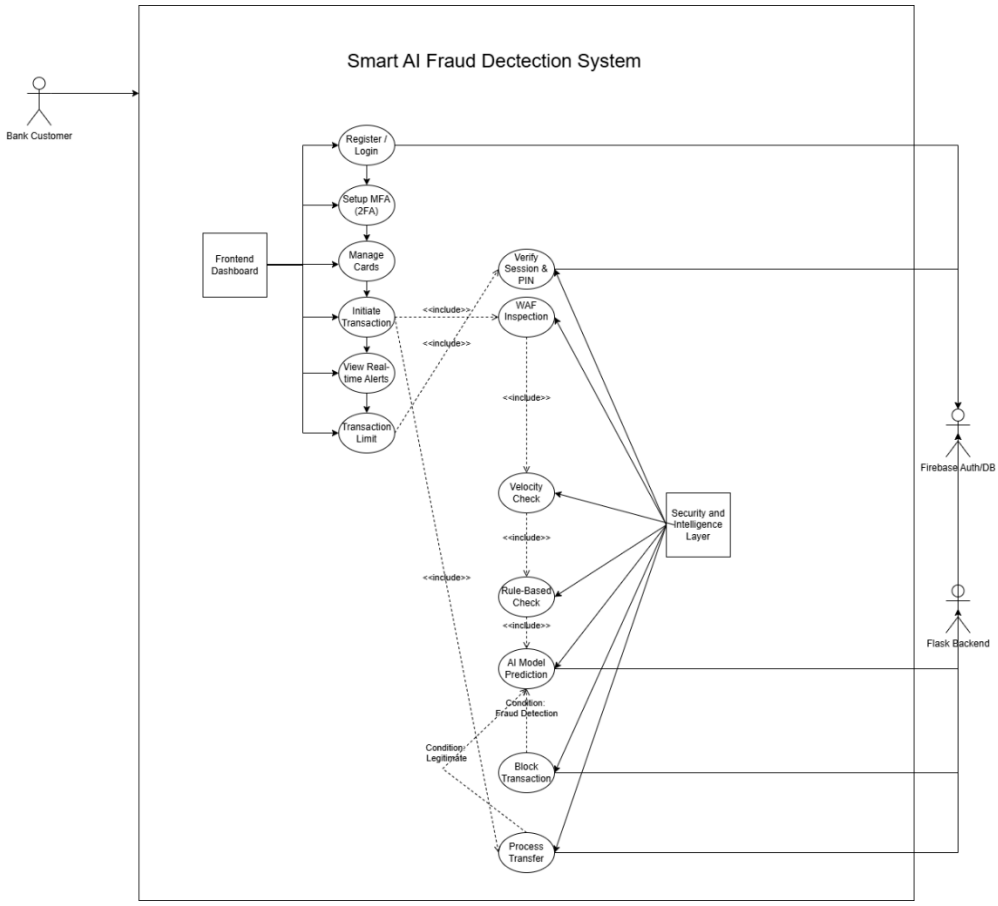

Figure 26 Use Case Diagram

The use case diagram demonstrates the functional requirements and interactions between the main parties (bank customer) and system elements (frontend, Firebase, and Flask backend). The system boundary involves the Intelligent AI Fraud Detection System that is architecturally organized to a frontend dashboard and a security and intelligence layer.

Primary Actor Interactions:

The system events are mostly started by the bank customer. The frontend allows customers to carry out account management functions like registration/ login, configuring MFA (2FA) and managing cards. Initiate Transaction is the most fundamental use case and it allows customers to conduct peer-to-peer (P2P) transfers

or bank top-ups. Firebase supports these operations and serves as a second-party that is involved in the process of authentication token management and user profile storage.

Security Dependencies (<<include>>):

In order to achieve the output of the Design Security approach, the diagram has the relationships of the form of the <<include> that imposes severe validation procedures. The first use case transaction involves the use of WAF, where all financial transactions are initially filtered by web application firewall against malicious content (e.g. SQL injection) before any processing can take place. The process will then include rate checks (to prevent automated bot attacks), and rule-based checks (to eliminate glaring anomalies e.g. amounts above $50,000). Such and such mandatory inclusion paths provide that no transactions go to the smart layer without first going through initial security gates.

Intelligence and Decision Logic (<<extend>>):

The AI model predicting use cases is an instance of core decision-making capability, which interacts with the Flask backend participant. This figure employs the relationship of conditional outcome of this prediction which is the relationship of <<extend>>. In case the AI model determines that a particular transaction is high-risk, the Block Transaction use case will continue the base flow and issue an alert log in Firebase. On the other hand, when the model considers the activity to be safe, the Process Transfer use case will stretch the flow to the funds transfer. This isolation of concern indicates the dynamism of the system in its ability to change behaviour according to real time inference outcomes.

Use Case Specification

4.2.1.1 Initiative Transactions

Field	Description
Use Case ID	UC-01
Use Case Name	Initiate Transaction
Primary Actor	Bank Customer
Secondary Actors	Flask Backend (AI Engine), Firebase (Database)
Description	The user performs a financial transfer (P2P or Top-Up). The system validates the request through security layers (WAF, Velocity) and uses AI to predict fraud before processing.
Pre-conditions	1. User must be logged in. 2. User must have a sufficient wallet balance.
Post-conditions	Success: Balance is updated in the database.Fail: Transaction is blocked and an alert is logged.
Main Flow	1. User selects transaction type and enters amount/recipient. 2. User clicks "Process Transaction". 3. System (WAF) scans for malicious payloads (SQLi/XSS). 4. System checks if user is spamming (Velocity Check). 5. System (Rule Engine) checks for high-value anomalies. 6. System (AI) predicts the transaction is Legitimate.

	7. System updates the database (deducts sender, adds to recipient). **8. System displays a "Success" message.**
Alternative Flows	**A. WAF Block (Security)** **3a. System detects malicious code.** **3b. System blocks request and displays "WAF Blocked".** **B. Fraud Detected6a. AI Model predicts Fraudulent.** **6b. System blocks the transfer.** **6c. System displays a Red "BLOCKED" alert.**

Table 2 Initial Transaction

4.2.1.2 Registration

Field	Description
Use Case ID	**UC-02**
Use Case Name	**Setup MFA (Two-Factor Authentication)**
Primary Actor	**Bank Customer**
Secondary Actors	**Firebase (Auth)**
Description	**The user enables 2FA by scanning a QR code to link their account with Google Authenticator.**
Pre-conditions	**User is logged in and MFA is currently disabled.**

Post-conditions	**mfaEnabled is set to TRUE in the user profile.**
Main Flow	**1. User clicks "Setup MFA".2. System generates a secret key and displays a QR Code.3. User scans the QR code with their mobile app.4. User enters the 6-digit code from the app.5. System verifies the code is correct.6. System enables MFA for the account.7. System displays "Setup Successful".**
Alternative Flows	**A. Invalid Code5a. System finds the code incorrect.5b. System displays "Invalid Code" error.5c. User tries again.**

Table 3 Registration

4.2.1.3 Manage Cards

Field	**Description**
Use Case ID	**UC-03**
Use Case Name	**Manage Cards (Tokenization)**
Primary Actor	**Bank Customer**
Secondary Actors	**Firebase (Database)**
Description	**The user adds a credit card. The system tokenizes the data (hiding the real number) and stores it for future use.**
Pre-conditions	**User is logged in.**
Post-conditions	**A new card token is saved to the database.**
Main Flow	**1. User enters card details (Number, CVV, Expiry).** **2. User clicks "Save Card".**

	3. System generates a secure Token (e.g., TOK_123). **4. System discards the raw card number.** **5. System saves the Token and last 4 digits.** **6. System updates the UI with the new card option.**
Alternative Flows	**A. Invalid Format** **2a. System detects invalid card number.** **2b. System shows error message.**

Table 4 Manage Cards

4.2.1.4 Transaction Limit

Item	**Description**
Use Case ID	**UC-06**
Use Case Name	**Set Transaction Limit**
Primary Actor	**Bank Customer**
Secondary Actors	**Firebase (Database)**
Description	**The user sets a maximum limit for outgoing transactions to prevent large unauthorized transfers.**
Pre-conditions	**User must be logged in.**
Post-conditions	**The transactionLimit field in the User Profile is updated.**
Main Flow	**1. User clicks the limit display on the dashboard.** **2. System opens the "Set Daily Limit" modal.** **3. User enters the new amount (e.g., 5000).** **4. User enters their 6-digit PIN for confirmation.**

	5. User clicks "Save Limit". **6. System verifies the PIN.** **7. System updates the limit in the database.** **8. System displays "Limit Updated" message.**
Alternative Flows	**A. Invalid PIN** **6a. System determines the PIN is incorrect.** **6b. System blocks the update and shows "Access Denied".**

Table 5 Transaction Limit

4.2.2 Data Flow Diagram (DFD)

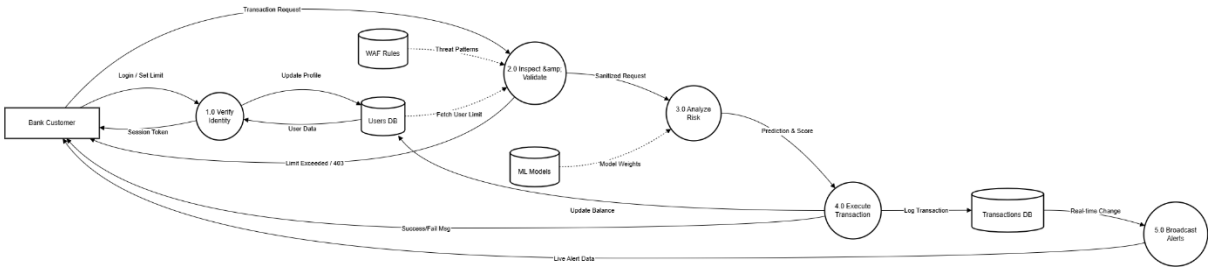

Figure 27 Data Flow Diagram

Overview of System Processes

Level 1 Data Flow Diagram represents the process of changing the data coming (bank customers) to the system into the form of external data (bank customers) contained in the storage layer. This architecture is broken down into five processes namely authentication, inspection of traffic, risk analysis, transactions and broadcasting of alerts. The processes are interacting with four types of data stores: the dynamic user and transaction database (Firebase), and the read-only storage of WAF rules and ML models.

Authentication and Access Control (Process 1.0)

A workflow, when a user provides login credentials to Process 1.0 (Authentication), takes off. This will access the Users DB and make a query to get the stored user profile and verify the credentials. After a successful authentication, the system sends a user a session token that creates a secure state during further interactions. This makes it possible that only authorized persons are allowed to make financial requests.

Security Screening and Risk Analysis (Processes 2.0 & 3.0)

In case an authenticated user is trying to make a transfer, a request message is directed to Process 2.0 (Traffic Inspection). This process is called to retrieve regular expression patterns stored in WAF Rules store to scan malicious payloads before any business logic is executed. In case a threat is identified, an error of 403 Forbidden is sent instantly to the user. The valid traffic is sent to Process 3.0 (Risk Analysis), as a processed request. This intelligent layer loads pre-trained model weights provided via the ML Model Store to execute inference, provides predictions and scores (e.g. Fraud: 95 percent) and forwards them to the execution phase.

Transaction Execution and Real-Time Feedback (Processes 4.0 & 5.0)

The decision made in the AI is implemented by Process 4.0 (Execute Transaction). When the transaction is valid, it does two important write operations: it makes an update to the account balance of the Users DB and permanently logs the transaction in the Transactions DB. It then sends out a success/failure message to the user. Lastly, to provide the transparency Process 5.0 (Broadcast Alerts) will check the Transactions DB on the appearance of any new records. When changes are detected (through real-time listeners), it updates the Live Alert Data back to the dashboard of the user and offers immediate visual feedback of the safe functioning of the system.

4.2.3 Database and Entity Relationship Design Overview

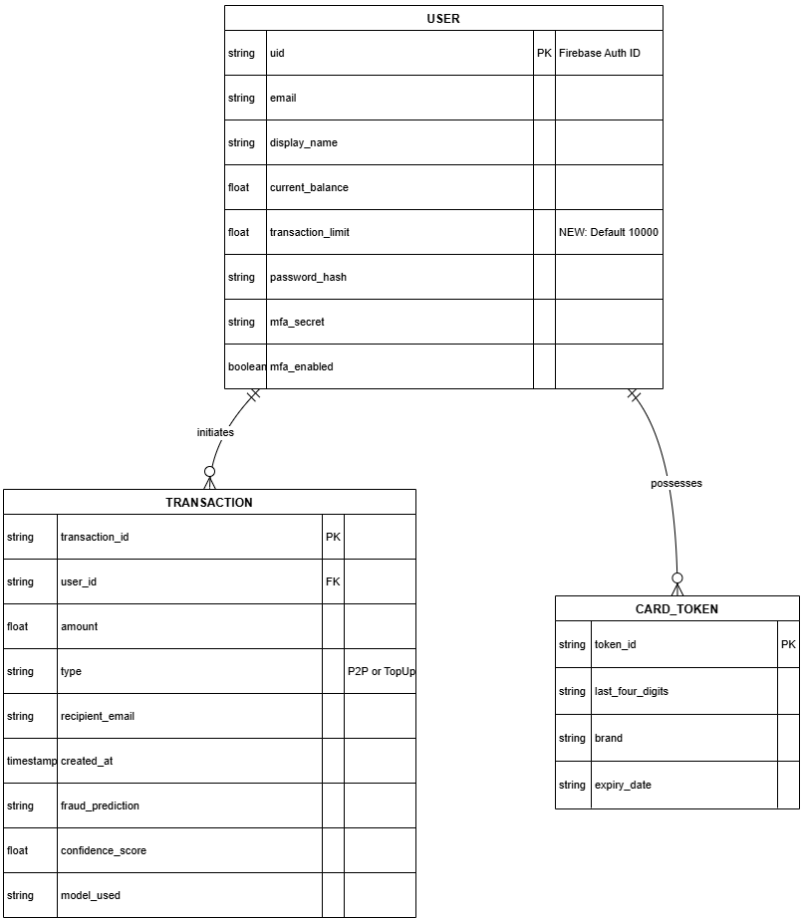

Figure 28 ERD Diagram

Core User Identity and Security

Entity-Relationship Diagram (ERD) represents graphical representation of the logical design of a database and the central element of the graph is the USER entity which is the major data object. The users are identified uniquely using a uid (primary key), which is a direct mapping of the Firebase Authentication service. In addition to normal profile data (email and display_name), User entity is provided with high-security capabilities. It comprises such attributes as mfa_secret and mfa_enabled to provide the required schema to accommodate the time-based one-time password (TOTP) logic needed by the multi-factor authentication module.

Transaction Logging and AI Integration

The system is highly restrictive in its one to many (1:N) relationship between User and TRANSACTION entities; this implies that any given user can trigger several financial activities. But every transaction should be associated with a valid account holder using the user id foreign key. Importantly, the Transaction entity is not merely a financial journal but also a log of analysis of fraud detecting engine. It directly stores the results of fraud prediction (legitimate or a fraud) and confidence score (a floating-point number in the range of 0 to 1). Such design will provide that every decision of the AI model will be stored permanently to be audited and retrained later on.

Payment Tokenization and Cardinality

In order to follow the best practices of PCI-DSS (Payment Card Industry Data Security Standard), the database does not store the sensitive financial information in user profiles, but rather it builds a special CARD_TOKEN record. This relationship is given as one-to-many (optional) that is; a user may have no or many payment methods. This entity will have just the secure tokenid and last four digits, which are displayed, so the original credit card number is not remotely available within the database. This identity disconnection, historical records of transactions, and payment validation guarantees a commercial and safe data infrastructure.

4.2.4 Flowchart

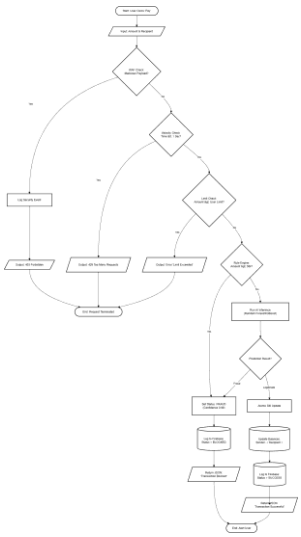

Figure 29 Flowchart

Phase 1: The security filtering (middleware)

The flowchart starts once a user makes a request in form of a transaction. The system filters the request before subjecting it to any business logic. Should the WAF identify any malicious behaviour: SQL injection strings or cross-site scripting tags, the request is without any further ado stopped and a 403 Forbidden error is generated. When the request is clean it goes through to the rate check. This module verifies the current timestamp against the last activity of the user; where time difference less than 1 second means that system blocks the request and sends a 429 Too Many Requests error to avoid automated bot attacks.

Phase 2: Hybrid Intelligence Analysis

When the request has been cleared by the security filter, it goes to the intelligence phase. In the first place, the deterministic rules engine checks the transaction against hard-coded rules (e.g. amounts over 50,000). Rules triggered result in the transaction being reported as high-confidence (0.99) fraud without any additional examination. Once there are no rules triggered, data is sent to the AI inference engine. The machine learning models (Random Forest/XGBoost) calculate the probability scores of legitimate and fraudulent transactions using the characteristics of the transaction.

Phase 3: Execution and Feedback

According to the final decision, the system uses one of the two routes. In case of fraud, the system will block any money transaction, store the incidence in the database with the status of blocked and alert the user. In case the transaction is genuine, the system makes an atomic database operation: it debits the funds of the sender and credit funds of the receiver on the same operation. Lastly, it logs a success status and communicates the frontend dashboard through a real-time listener.

4.3 Database Design

Since the financial data can be of a variety of types, such as structured profiles of users and great amounts of transactions, we chose a NoSQL document-oriented database. This schema has an optimisation of read-intensive operations, and it accommodates real-time analytics dashboards. The database contains two major collections:

4.3.1Users Table

1. Users Collection (users):

COLLECTION	FIELD NAME	DESCRIPTION	DATA TYPE	CONSTRAINT
users	uid	Unique Authentication ID provided by Firebase Auth.	string	Primary Key (PK)
	email	User's registered email address.	string	Unique, Not Null
	displayName	User's full name for interface display.	string	Not Null
	balance	Current available account balance.	number (float)	Default 0.00, Min 0
	transactionLimit	User-defined daily maximum for outgoing transfers.	number (float)	Default 10000
	mfaEnabled	Status flag indicating if 2FA is active.	boolean	Default false
	mfaSecret	Base32 secret key used for generating TOTP codes.	string	Encrypted
	pinHash	Securely hashed 6-digit PIN for sensitive actions.	string (hash)	Not Null

Table 6 User Table

The user collection is the identity repository of the application, which is directly connected to Firebase Authentication by the primary key of the uID. The design will provide a one-to-one correspondence between user identity authentication tokens and user profiles stored. Critical security information is confined in this document: the encrypted hash of user 6-digit security PIN, rather than of the actual password, is stored in pinHash, and cannot be reverse engineered

by even database administrators. Also, the transactionLimit field underlines the risk management module of the system, allowing the personalization of daily spending limits, which the backend will check and confirm the transfer of funds before any transactions are made.

4.3.2 Transaction Table

2. Transaction Collections:

COLLECTION	FIELD NAME	DESCRIPTION	DATA TYPE	CONSTRAINT
transactions	(Doc ID)	Auto-generated unique transaction reference ID.	string	Primary Key (PK)
	userId	Reference to the user who initiated the transaction.	string	Foreign Key (FK) ref users.uid
	amount	The numerical value of the financial transaction.	number (float)	> 0, Not Null
	type	The category of transaction (e.g., "p2p", "topup").	string	Not Null
	recipient	Email or identifier of the transaction receiver.	string	Not Null
	timestamp	Server-side time when the transaction occurred.	timestamp	Server Timestamp
	status	Final outcome (e.g., "SUCCESS", "BLOCKED_AI", "BLOCKED_WAF").	string	Not Null
	aiResult	Nested map containing AI inference details (prediction, confidence score, model used).	map (object)	Nullable (if blocked by WAF)

Table 7 Transaction Table

These sets of transactions act as an audit trails which are permanent in nature and all transactions of financial nature are recorded in the system. This NoSQL document structure, unlike a normal relational table, is very efficient in complex, nested data especially the aiResult map. This area is paramount to the explainable AI prerequisites of the system, since it does not just safeguard the ultimate finding (condition) but also the exact model executed (e.g., “Random Forest Expert”) and the trust score created at the time thereof. The user id is used as a foreign key which allows the system to easily query and rebuild the complete financial history of the user to generate real-time streams of activity without going through irrelevant record entries.

4.3.3 Payment Method Table

3. Payment Method Collection:

COLLECTION	FIELD NAME	DESCRIPTION	DATA TYPE	CONSTRAINT
payment_methods	tokenId	Unique secure token representing the card details.	string	Primary Key (PK)
	userId	Reference to the card owner.	string	Foreign Key (FK) ref users.uid
	brand	Card network (e.g., "Visa", "MasterCard").	string	Not Null
	last4	The last four digits of the card number for identification.	string (fixed 4)	Not Null
	expMonth	Card expiration month.	number	1-12
	expYear	Card expiration year.	number	Not Past Year
	Note	Raw card numbers and CVV codes are explicitly excluded		

		from storage in compliance with PCI-DSS standards.		

Table 8 Payment Method

The paymentMethods collection is aimed at providing financial transactions with ease of operation and fully complying with the requirements of the PCI-DSS (Payment Card Industry Data Security Standard). This collection is not based on a tokenization model as in the case of simple tables that could hold raw information. The primary key tokenId is a secure reference string (e.g. tok_123) allocated by the payment gateway, to act as a replacement of real sensitive credentials. The system retains no sensitive data other than the non-sensitive metadata of the brand (e.g. Visa/Mastercard) and the final four digits of the card to provide user experience without making security a consideration. This design also can allow the frontend to show recognizable card options to users (e.g., Visa card ending in 4242) and at the same time, even in case of database breach, actionable financial information (full primary account number or CVV/CVC code) will not be available to the attackers. The field of the type of user ID is a foreign key, which creates a one-to-many relationship whereby a single customer can safely use numerous payment tools.

4.4 Interface Design

The user interface is carefully modelled to accommodate high user-friendliness but without compromising on high functionality to be able to address industry requirements whereby fraud reporting must be transparent.

4.4.1 MFA Interface

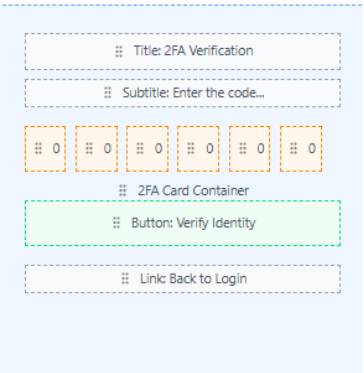

Figure 30 MFA Interface

It has access control interface where registration and login are divided to keep the user errors to a minimum. There is a special modal that is used in the process of 2FA. It is a dynamic QR code generator which is based on otpauth library and enables individual users to check their identity through external authentication apps. This modal offers visual feedback immediately and then shuts itself and unlocks the dashboard as the code is verified.

4.4.2 Transaction Hub Interface

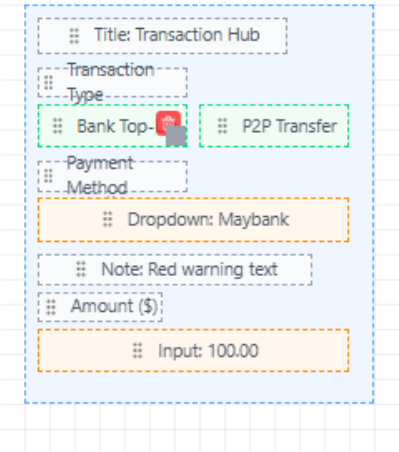

Figure 31Transaction Hub

The main transaction form applies dynamic DOM manipulation to accommodate the input of the user. With the option of P2P transfer being selected, the interface shows the field of Recipient Email; whereas the option of Bank Top-up shows nothing and displays the field of Payment Method. This conditional rendering helps to decrease the confusion and direct users on the conditions of a type of transaction.

4.4.3 Live Activity Interface

Figure 32 Live Activity Interface

A Live Activity Feed was introduced in order to visualize the velocity of systems through the onSnapshot listener of Firestore. This element makes the transaction cards color coded: green means successful transfer, red means blocked fraud attempt. The cards indicate the trust score

and the AI model that made the decision, which directly answers the issue of the interviewee: lack of transparency (Siew, 2025).

4.4.5 Login Interface

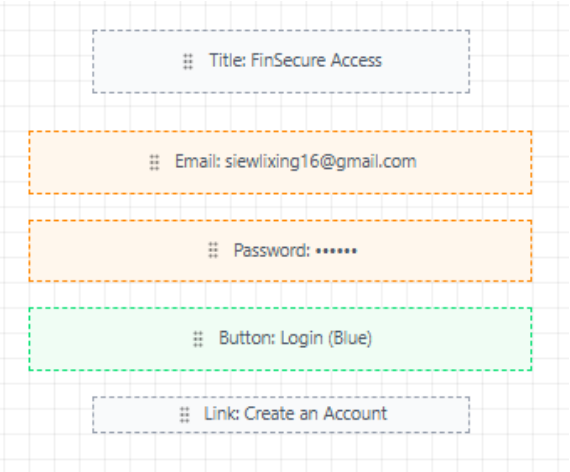

Figure 33 Login Interface

This is a wireframe that shows the initial layout of the FinSecure Access authentication module which is the safe gateway of the fraud detection system. The interface is designed with the vertical stacking layout, which favours clarity and minimum cognitive load of the users. It contains two major input fields, email and password, which are explicit enough to instruct the end users on keying in the credentials. There is a conspicuous button of log-in (underlined in blue) positioned directly underneath the inputs to provide a clear call-to-action, and the button of creating an account is placed at the bottom to provide another means of navigation without interfering with the overall working process. The design will make the authentication process to be user-friendly and security-oriented.

4.4.6 Setup MFA Interface

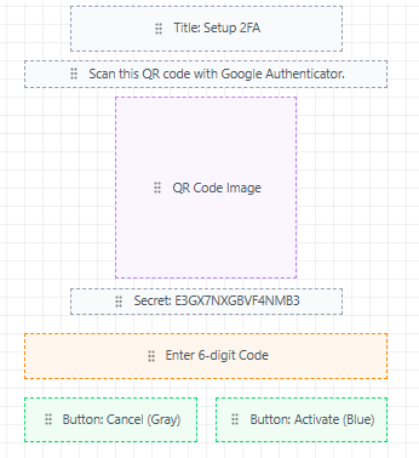

Figure 34 Setup MFA Interface

The wire frame represents the interface of multi-factor authentication (MFA), which is an essential security component to be used to execute two factor authentication (2FA). The trend revolves around dynamically generated QR code, which is the main means of connecting user accounts to time-based one-time password (TOTP) applications, such as Google Authenticator. Just under the visual code, there is a text key that is given as a backup entry feature. Before allowing this option, the interface has a special input box where a 6-digit code is to be entered to check the synchronization. To ensure an action button hierarchy, the main action button, which is the Activate button, has been set to blue to indicate completion, and the secondary one, which is the cancel button, is styled neutrally to lessen distraction.

4.4.7 Manages Card Interface

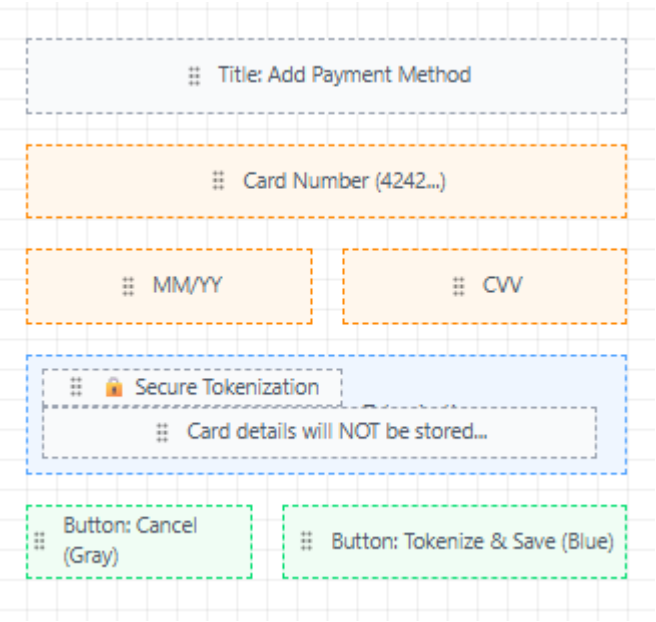

Figure 35 Manage Card Interface

The wireframe depicts a financial instruments onboarding module, the Add Payment Method, which is created to promote secure onboarding of financial instruments. The interface adheres to the conventional practices of the banking industry, where it asks to enter the card number, expiration date (MM/YY) and CVV, and then displayed in a traditional grid format, to reduce the error of the user. One of the main elements of this design is the specific lock icon and explicit Secure Tokenization disaster with a lock icon on the page. This image helps strengthen the privacy first architecture of the system (PCI-DSS compliant) by making the users aware that card information is not stored in the database in the first place. Their main action button is clearly named as Tokenize & Save (in blue) as opposed to the generic one, which will give technical visibility that sensitive data will be transformed to secure tokens and then stored.

4.4.8 Transaction Limit Setup

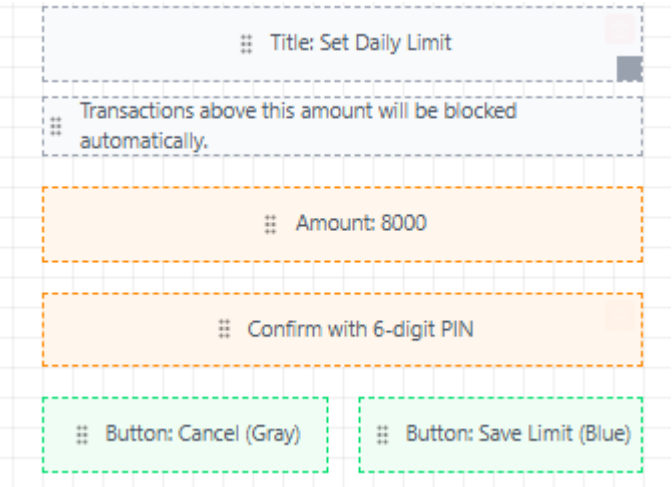

Figure 36 Transaction Limit Setup Interface

The wireframe illustrates the Set Daily Limit module of configuration which is a user-friendly Risk management feature, where customer can give their maximum thresholds on transactions made. The user-face is presented in the form of a narrow-scoped modal window to avoid distraction out of the most urgent activity. Its main entry point is the limit amount (say 8000), and clearly a custom explanatory label saying that transactions above this amount would be blocked automatically. In a bid to achieve design security, the module has a mandatory field of Confirm 6-digit PIN. This is necessary to ensure that security settings are not changed by unauthorized users who have physical access to the device to which the logins are made. The workflow is closed by the primary blue button that is named Save Limit to save the changes to the database, and a secondary Cancel button.

4.5 Execution

The implementation stage includes a combination of different technologies to form an uninterrupted hybrid defence pipeline. Processing logic has a rigid course of operation:

4.5.1 Technology Stack Selection

- **Backend:**

```
import joblib
from flask import Flask, request, jsonify, send_from_directory
from flask_cors import CORS
import numpy as np
import pandas as pd
import time
from sklearn.preprocessing import StandardScaler

app = Flask(__name__)
CORS(app)
```

Figure 37 Backend Architecture

The backend architecture is created using Flask micro framework, as it is a lightweight framework and is particularly effective in processing the HTTP requests. The system incorporates important data science libraries including Pandas and NumPy to perform high-performance numerical programs and structure data as shown in Figure 3.x. Importantly, the Joblib library imports and loads trained machine learning models (modelcreditcard.joblib and modelpaysim.joblib), and it can infer in real-time on transaction data that is entered. Assessment of Flask-CORS also provides a secure cross-origin resource sharing to enable the frontend to interact with the Python API without security conflicts by the browser.

- **Frontend:**

```
!-- Firebase SDK Logic -->
script type="module">
  import { initializeApp } from "https://www.gstatic.com/firebasejs/11.6.1/firebase-app.js";
  import { getAuth, onAuthStateChanged, createUserWithEmailAndPassword, signInWithEmailAndPassword,
  import { getFirestore, doc, setDoc, getDoc, collection, query, where, onSnapshot, runTransaction,
```

Figure 38 Client Application

Client application The client application has a smart frontend architecture with Firebase Modular SDK (v11). As Figure 3.x shows, the system directly imports particular modules initializeAPP, getAuth, and getFirestore instead of dynamic loading the library. This pruning

method is best in optimization of loading times and performance. The technology stack using Google Firestore, a NoSQL cloud database, is used to synchronize real-time information. This facilitates real time update of transaction logs and user balances in dashboards without manually page revision that would result in responsive and a seamless user experience.

4.5.2 Rule-Based Evaluation

```
try:
    # Rule Engine
    if 9000 <= amount < 10000:
        return jsonify({'prediction': 'Fraudulent', 'confidence': 0.85, 'model_used': 'Rule-Engine (Structuring Detect)'})

    if amount > 50000:
        return jsonify({'prediction': 'Fraudulent', 'confidence': 0.99, 'model_used': 'Rule-Engine (High Value)'})
```

Figure 39 Rule Amount and Limitation

It is a code snippet that is used to implement a deterministic rules engine, which serves as a high-ranking filtering layer prior to or in conjunction with machine learning models. Its reasoning specifically aims at two financial crime patterns. First, there is the condition 9000 < Amount 10000 which identifies the tactic of structuring (or evading detection by) criminals choosing to maintain amounts under the normal 10,000 reporting limit. Such transactions are automatically flagged within a medium level of confidence of 0.85 on the system. Second, the condition amount > 50000 is used as a strict security control of large money deals that surely warn as fraudulent and are easily avoided to suffer considerable losses (confidence 0.99). The system provides the front end with the accurate and interpretable feedback to users or compliance officers by generating certain model used strings (e.g., Rule Engine (High Value)).

4.5.3 AI Model Inference

```
FEATURES_CREDITCARD = [
    'V1', 'V2', 'V3', 'V4', 'V5', 'V6', 'V7', 'V8', 'V9', 'V10',
    'V11', 'V12', 'V13', 'V14', 'V15', 'V16', 'V17', 'V18', 'V19', 'V20',
    'V21', 'V22', 'V23', 'V24', 'V25', 'V26', 'V27', 'V28',
    'Scaled_Amount', 'Scaled_Time', 'Hour'
]

FEATURES_PAYSIM = [
    'amount', 'oldbalanceOrg', 'newbalanceOrig', 'oldbalanceDest', 'newbalanceDest',
    'amount_gt_oldbalanceOrg', 'newbalanceOrig_is_zero', 'oldbalanceDest_is_zero',
    'type_CASH_IN', 'type_CASH_OUT', 'type_DEBIT', 'type_PAYMENT', 'type_TRANSFER'
]

NUMERICAL_FEATURES_PAYSIM = [
    'amount', 'oldbalanceOrg', 'newbalanceOrig', 'oldbalanceDest', 'newbalanceDest'
]
```

Figure 40 AI Features

The given piece of code emphasizes the feature engineering phase, which is one of the most important parts of the machine learning process that converts raw data into the format that can be used to infer the model. The system clearly outlines two separate sets of features in order to process various types of transactions. The feature list used to complete credit card transactions is FEATURES_CREDITCARD that uses variables based on the Principal Component Analysis(PCA)(V1 - V28) in addition to time related features such as hour which make it compatible with the layout of the Kaggle Credit Card Fraud dataset. On the other hand, in the case of mobile money transfers, the behavior-based attributes such as oldbalanceOrg (balance of sender before transaction) and newbalanceDest (balance of recipient after transaction) are selected in the list of features called FEATURES-PAYSIM. This distinction will make sure that the Random Forest and XGBoost classifiers are supplied with domain-specific data and are as accurate as possible to recognise anomalies like account hijacking or money laundering.

4.5.4 Response & Atomic Updates

```
if (type === 'credit_card') { // Top Up
    await runTransaction(db, async (t) => {
        const ref = doc(db, "users", auth.currentUser.uid);
        const bal = (await t.get(ref)).data().balance;
        t.update(ref, { balance: bal + amt });
    });
```

Figure 41 Atomic Operation

The outcome is the end result which is in form of a JSON object. On the client-side, when the prediction is that it is legitimate, a Firebase transaction is made. This atomic operation makes sure that both the sender and the recipient balance is debited and credited respectively and in case of failure of any of the two operations, the whole transaction is reversed to avoid data inconsistencies.

4.5.5 Transaction Limit

```
// Default Limit
if (!userData.transactionLimit) userData.  (property) HTMLElement.innerText: string
document.getElementById('limit-display').innerText = `Limit: $${userData.transactionLimit.toLocaleString()}`;
```

Figure 42 Transaction Limit

The snippet of the code shows the client-side setup process of the feature of transaction limit. When loading the user profile, a conditional check (if (!userData.transactionLimit) is carried out by the system to make a decision whether a custom limit was set before. Whenever the value in the database is not set (typically new users or legacy users), the system will automatically set the default security threshold (which is usually set to $10,000) to provide instant protection to the account. The code then goes on to address the limit-display element of the DOM, and changes the innerText to show the active limit. Most importantly, it makes use of the.toLocaleString() to decompose the numeric value into the correct comma separator (e.g., 10000 to 10, 000) to improve the readability and user experience of the dashboard.

4.5.6 WAF Middleware

```python
@app.before_request
def waf_middleware():
    """
    Mini-WAF to intercept and inspect all incoming requests for malicious payloads.
    """
    # 1. Helper Function to scan a string
    def scan_payload(value, source):
        if not isinstance(value, str):
            value = str(value)
        for threat_type, patterns in WAF_RULES.items():
            for pattern in patterns:
                if re.search(pattern, value):
                    print(f"!!! WAF BLOCKED: {threat_type} detected in {source} (Payload: {value}) !!
                    return threat_type
        return None
```

Figure 43 WAF Middleware

The code snippet shows how it is possible to implement a Web Application Firewall (WAF) middleware using the Flask framework in Python. Using the decorator of the name before request this is hooked into the global request life cycle and is therefore executed by every incoming HTTP request before any business logic has been run. The main principle is based on a helper procedure, which is scan payload, that scans a dictionary of constant regular expressions (WAF_RULES). It sequentially searches the incoming data stream by the method research to locate any known attack signature that may be SQL injection keywords or cross-site scripting (XSS) tags. In case a pattern match is detected, the system detects the particular type of threat and ends the request instantly, which in effect will counter threats at the network edge.

4.5.7Velocity Checking

```
# Velocity Check
if user_id in last_action_time:
    if current_time - last_action_time[user_id] < 1:
        return jsonify({
            'prediction': 'Fraudulent',
            'confidence': 1.0,
            'model_used': 'Security Gateway: Velocity Check',
            'error': 'Action too frequent. Please wait 1 second.'
        }), 429
last_action_time[user_id] = current_time
```

Figure 44 Velocity Checking

The code snippet shows how the Velocity Tracker which is an essential security control that is meant to curb the automated bot attacks and transaction flooding is implemented. The system maintains an in-memory Python dictionary of last-action-time, which is used to store the exact time of the last activity of each active user-id. When a new transaction request is received the system compares the time difference between the current time of the system and the timestamps stored in it. Provided that this period is lower than a specific 1-second threshold, the request is thus instantaneously considered as an anomalous behavior. This system totally disregards the AI model, and it returns a 429 Too Many Requests HTTP error with a prediction of 429 with a confidence of 1.0, which is a fraud, successfully preventing attacks of high frequency before they overload the system resources.

4.6 Screenshot

4.6.1 Dashboard

- **Figure 4.1: The Authentication & MFA Interface.**

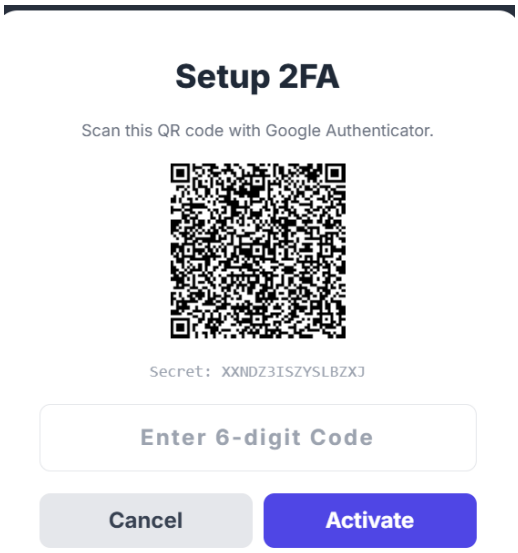

Figure 45 MFA Setup

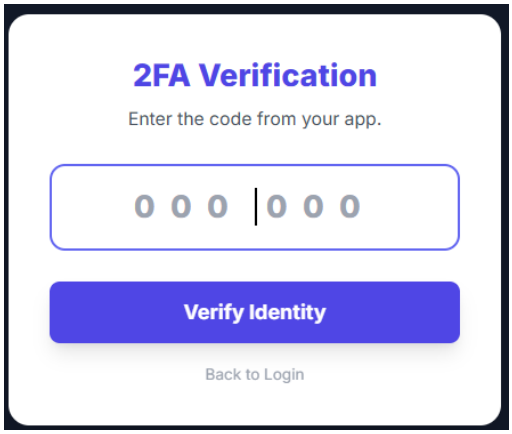

Figure 46 MFA Verification

The interface of MFA setup is built to be more of a centralized modal overlay, and it is built with the focus on facilitating a high level of security through the means of being easy to navigate. To establish immediate synchronization with mobile authentication applications, the visual hierarchy focuses on dynamically created Quick Response (QR) code that encodes the personal key of the user in Terms of TOTP. To be aware of the possible hardware shortcomings, the interface integrates a failover system: the initial alphanumeric key (i.e., E3GX...) is presented directly below the graphic to be entered manually. The workflow includes a check of

possession: users have to provide a valid 6-digit kind of token in the field that verifies the user, only after that the primary Activate button (rendered in high-contrast blue) can be pressed and the secure link is verified, only then account policies are applied.

Figure 4.2: The Main Dashboard & Transaction Hub.

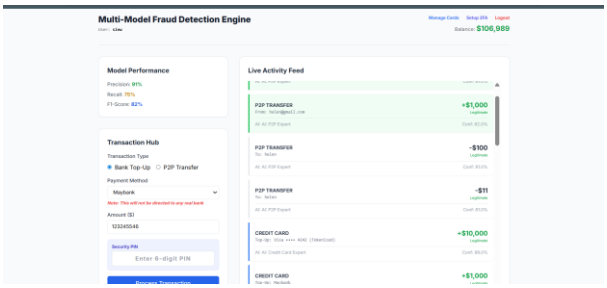

Figure 47 Dashboard

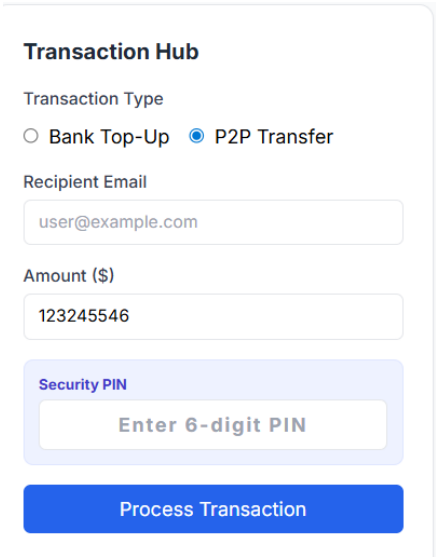

Figure 48 Transaction Hub

The main working centre of the user is the so-called "Transaction Hub" with its dynamic shape depending on the situation. Transaction types are selected by radio buttons, which are used to toggle transactions between P2P Transfer and Bank Top-up types, causing the instantaneous update of the interface: When changing to P2P Transfer, the Recipient Email field is shown, and when changing to Bank Top-up, the card selection options are shown in place of the email field. This design is the least cluttered as it shows only the fields which are relevant. In order to provide the integrity of the transactions, the workflow will impose a Security PIN challenge, and then the Process Transaction button will be active. This enhanced authentication system

does not allow unauthorized users to make financial transfer without secondary credentials even when they are in an active session.

Figure 4.3: Fraud Detection Alert.

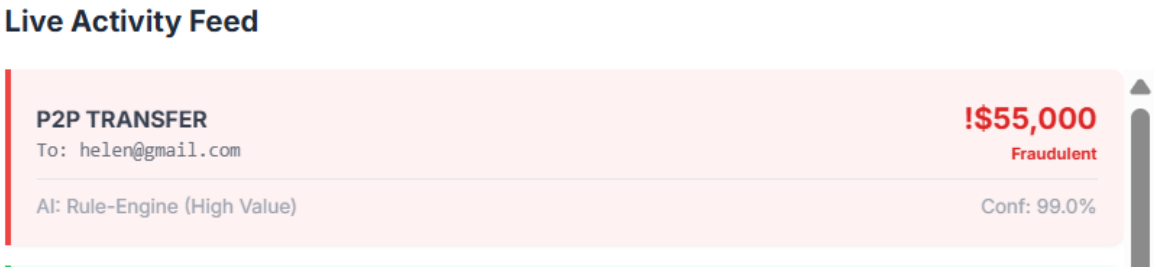

Figure 49 Fraud Detection Alert

This screenshot shows that the system can give instant visual feedback when it realizes that there is something malicious. This is an alert card that uses a special strategy of semantic color-coding, where a red border and background is used to highlight to the user immediately that there is a critical security event occurring (blocked status). The interface also follows principles of explainable AI (XAI) by not only the underlying decision logic being visible, but the decision maker given in the interface is also (AI: Rule Engine (High Value)) and the probability score (Conf: 99.0) is visible. Such openness will make users properly comprehend the reason why a high-value transaction of $55, 000 was intercepted, which will strengthen faith in automated defences.

Figure 4.4: Live Activity Feed.

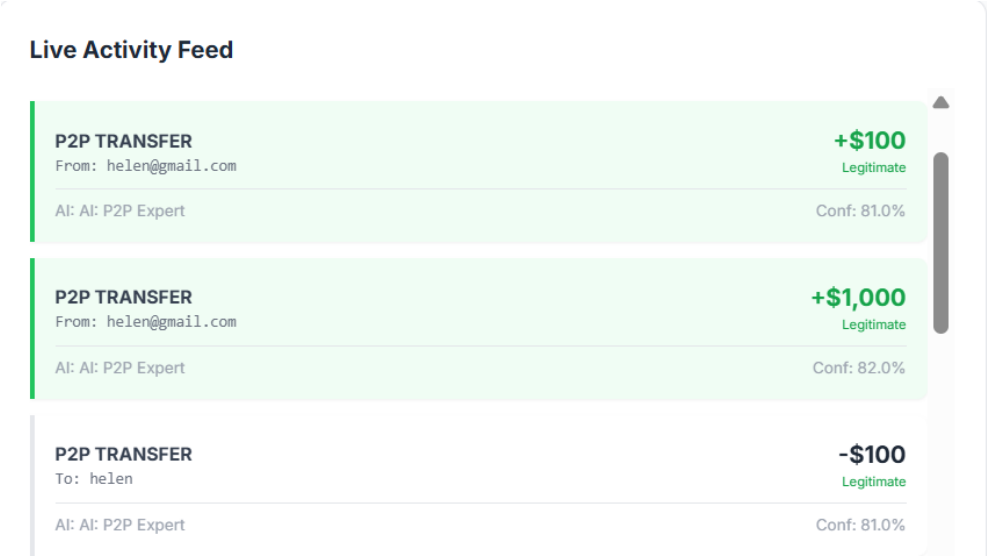

Figure 50Live Activity Feed

This screenshot shows the Live Activity Feed module which gives the user a clear audit trail. In contrast to alert statuses, valid transactions are shown with green semantic themes (borders and text) and are used to indicate the status of the transaction as legitimate, and the positive user feedback is reinforced. This interface shows that the system has the granularity of logging successful transmissions, but even in successful transmissions it clearly shows the AI confidence score (e.g., "Trust: 81.0%") and the model that was used (e.g., AI: P2P Expert). This is the design aspect that proves the fraud detection engine to be active on all the transactions, not only blocked transactions, which gives the user a continuous confidence that his financial operations are being tracked and confirmed as genuine and in real-time.

Figure 4.4: WAF Blocking Event

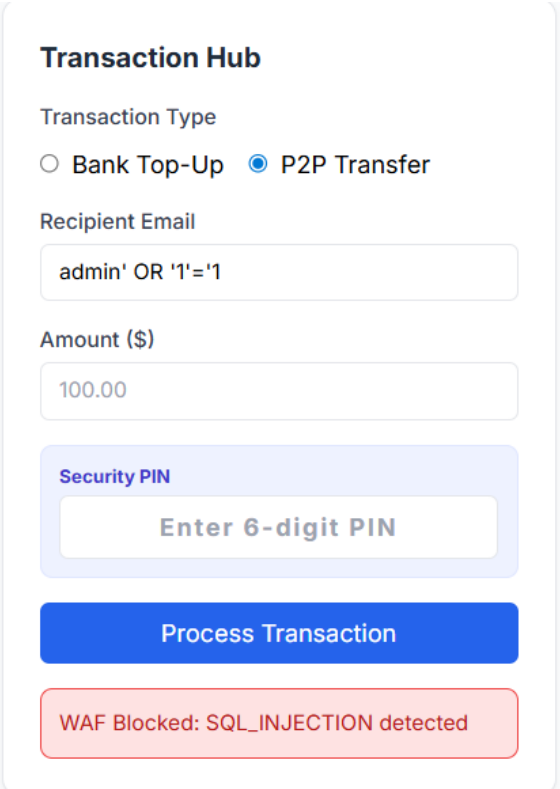

Figure 51WAF Blocking Event

The occurrence of this screenshot justifies the active security perimeter of the system in that Web Application Firewall (WAF) reacts to malicious input. The test scenario used a generic SQL injection query (admin' OR 1=1) to deliberately key in into the recipient email box with the intention of trying to gain unauthorized access to the database. The middleware layer of the system captures the request prior to getting to core processing logic. This interface also displays immediate blocking feedback through a high-profile red warning that reads WAF Blocked: SQL_INJECTION detected, is enough confirmation that expensiveness pattern matching regulations outset to the back entice managed to block malicious syntax out of user input.

Figure 4.5 Transaction Limit

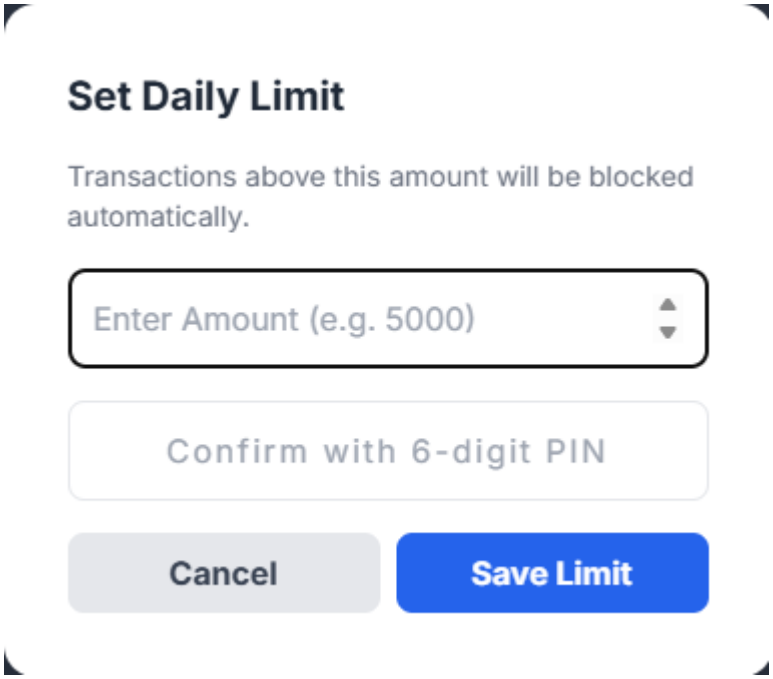

Figure 52 Transaction Limit

The screenshot presents a high-fidelity version of the pattern of the Set Daily Limit, which can change the original wireframes into working security controls. This interface gives the users the ability to have control over the risk management in a proactive manner where they can set limits on the maximum amount of money they can transact on a day-to-day basis. One of the security measures that are introduced in this module is defined as high authentication levels: the field with the 6-digit PIN is obligatory. This guarantees that sensitive security settings are not modified by non-participant persons in an active session in case they access it. The visual hierarchy is also applied in the visual design where a wide blue button (Save Limit) is used to direct the user to the primary action, with a grey button (Cancel) positioned at the neutral position in order to avoid submissions by mistake.

Figure 4.5 Manage Card

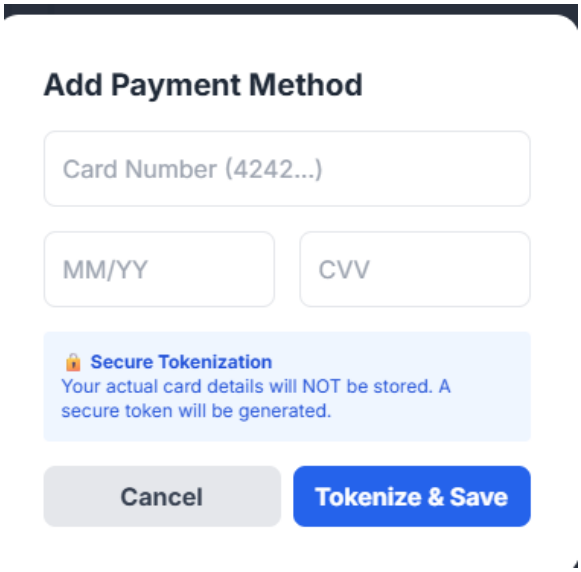

Figure 53 Manage Card

The screenshot presents a high-fidelity demonstration of the Add Payment Method module, which is developed according to the strict financial security requirements. The interface is used to enter vital information in the form of cards (number, expiration, CVV) in a clean and well-organized interface. One very important aspect of this implementation is the presence of a noticeable notice of the process of securing tokenization (marked in a light blue info box with a lock icon), only to clearly remind the users that their original data will not be retained. The main button is clearly named Tokenize and Save, which, again,, is in line with the system architecture in which sensitive data gets translated into secure tokens (i.e., TOK_123) prior to being sent to the database in line with the best practices of the PCI-DSS.

4.7 Summary

This chapter elaborates about how an intelligent fraud detection engine was successfully designed and implemented. The system architecture partially makes good use of a hybrid solution, with the speed of rule-based checks and the flexibility of machine learning models. The database structure is optimized with speed in terms of absorbing data and security in terms of tokenization. The interface design provides usability by means of MFA and real-time feedback feedback loops, as well as maintaining high standards of security. The proper combination of these elements has resulted in a strong prototype that has the ability to identify and prevent the simulated financial fraud in real time, which meets the main technical goals of this project.

CHAPTER 5: RESULT AND DISCUSSION

5.1 Introduction

In this chapter, the intelligent fraud detection engine is critically assessed in terms of both technical and operating stability and viability in a simulated fintech environment. The security protocols are the most important aspect in the financial technology industry, and hence, the ultimate goal of this testing stage is to prove the efficiency of the hybrid defense infrastructure suggested in the methodology. This architecture combines a Web Application Firewall (WAF), deterministic rule engine, and machine learning models in order to provide multi-layered protection against financial fraud. The evaluation plan utilized an effective hybrid methodology, which integrates both functional black-box testing to determine the integrity of the backend security controls in accordance with simulated cyberattacks, as well as user acceptance testing (UAT) to determine the efficiency of the interface of the system as well as user experience to the target user control group. By isolating these testing variables, the project was going to prove that the system does not only detect fraud accurately but also does not affect the smooth user experience needed to make the system acceptable.

5.2 Testing Design and Plan

The test plan was split into two phases to cover all the elements of the system:

5.2.1 Black Box Testing Strategy (Security Verification)

The black-box testing has been chosen to ensure that the security layers of the system are functional and do not examine internal code structures to achieve that. The test plan was mainly on API endpoints that were established in the application backend. In this approach, boundary value analysis and error guessing were used to test validation logic using stress. There were certain test cases:

5.2.1.1 Black Box Testing- Transaction Processing System

ITEM	DESCRIPTION
TEST CASE ID	**BB-01**

MODULE TESTED	**Transaction Processing System**
TESTING TECHNIQUE	**Equivalence Partitioning**
TEST DESCRIPTION	**Verify that a valid P2P transfer request with correct inputs results in a successful transaction and balance update.**
TEST STEPS	**1. Enter valid Recipient Email.** **2. Enter valid Amount ($100).** **3. Enter correct PIN.** **4. Click Process Transaction.**
TEST DATA	**Amount: $100PIN: Correct (123456)**
EXPECTED RESULT	**System displays "Success" message. User balance decreases by $100. Feed updates with a green "Legitimate" entry.**
ACTUAL RESULT	**Message "Success!" displayed. Balance deducted. Feed updated.**
STATUS	**Pass**

Table 9 Transaction Processing System

5.2.1.2 Black Box Testing- Transaction Limit Control

ITEM	**DESCRIPTION**
TEST CASE ID	**BB-02**
MODULE TESTED	**Transaction Limit Control**
TESTING TECHNIQUE	**Boundary Value Analysis**
TEST DESCRIPTION	**Verify system behavior at the exact boundaries of the user's daily limit (Limit set to $5,000).**

TEST STEPS	**1. Attempt transfer of $5,000 (Exact Boundary).** **2. Attempt transfer of $5,001 (Just Above Boundary).**
TEST DATA	**Limit: $5,000Input A: $5,000Input B: $5,001**
EXPECTED RESULT	**Input A: Transaction Allowed.** **Input B: Transaction Blocked with Limit Exceeded error.**
ACTUAL RESULT	**Input A proceeded successfully. Input B triggered the error popup.**
STATUS	**Pass**

Table 10 Transaction Limit Control

5.2.1.3 Black Box Testing- Input Validation

ITEM	**DESCRIPTION**
TEST CASE ID	**BB-03**
MODULE TESTED	**Input Validation (WAF)**
TESTING TECHNIQUE	**Error Guessing / Syntax Testing**
TEST DESCRIPTION	**Verify that the application correctly handles and rejects invalid characters or SQL syntax in text fields.**
TEST STEPS	**1. Navigate to Transaction Hub.** **2. In Recipient Email, enter a SQL injection string instead of an email.** **3. Click Process.**
TEST DATA	**Input: admin' --**

EXPECTED RESULT	**The system must not crash or execute the query. It must return a validation error or security block message.**
ACTUAL RESULT	**System returned "WAF Blocked" alert. No database error occurred.**
STATUS	**Pass**

Table 11 Input Validation

5.2.2 User Acceptance Testing (UAT)

User Acceptance Test (UAT) aims at verifying system usability as viewed by non-technical end-user. A sample group that represents the potential banking customers evaluates the dashboard interface. The test situation will involve the participants undertaking a sequence of explicit tasks:

5.2.2.1 User Acceptance Testing- Secure Fund Transfer

ITEM	DESCRIPTION
TEST CASE ID	**UAT-01**
FEATURE TESTED	**Secure Fund Transfer (Fraud Detection)**
TESTING TECHNIQUE	**User Acceptance Testing (UAT)**
TEST DESCRIPTION	**Verify that the user receives an immediate warning block when attempting a transaction that fits a high-risk fraud profile.**
TEST STEPS	**1. User login to the dashboard.** **2. User selects "P2P Transfer".**

	3. User inputs a high amount ($55,000) to an unknown recipient.4. User clicks "Process Transaction".
TEST DATA	**Amount: $55,000Recipient: stranger@gmail.com**
EXPECTED RESULT	**The system should display a Red Alert Sign stating "Fraudulent". The money should not be deducted from the balance.**
ACTUAL RESULT	**Red alert appeared. Balance remained unchanged.**
STATUS	**Pass**

Table 12 Secure Fund Transfer

5.2.2.2 User Acceptance Testing- Risk Management Control

ITEM	**DESCRIPTION**
TEST CASE ID	**UAT-02**
FEATURE TESTED	**Risk Management Controls (Daily Limit)**
TESTING TECHNIQUE	**User Acceptance Testing (UAT)**
TEST DESCRIPTION	**Verify that the user can successfully set a personal daily limit and that the system enforces it during a transfer.**
TEST STEPS	**1. User opens "Set Daily Limit" modal.** **2. User enters "$500" and confirms with their PIN.** **3. System confirms update.** **4. User attempts to transfer $600.**

TEST DATA	**Limit Set: $500Transfer Attempt: $600**
EXPECTED RESULT	**The system should successfully save the limit. The subsequent transfer of $600 should fail with a message: "Amount exceeds daily limit".**
ACTUAL RESULT	**Limit saved successfully. Transfer blocked with correct error message.**
STATUS	**Pass**

Table 13 Risk Management Control

5.2.2.3 User Acceptance Testing- Account Security

ITEM	**DESCRIPTION**
TEST CASE ID	**UAT-03**
FEATURE TESTED	**Account Security (MFA Login)**
TESTING TECHNIQUE	**User Acceptance Testing (UAT)**
TEST DESCRIPTION	**Verify that a user with MFA enabled cannot access the dashboard without providing the correct 6-digit verification code.**
TEST STEPS	**1. User enters email and password on the login screen.** **2. System prompts for "2FA Verification".** **3. User enters the code from their Google Authenticator app.** **4. User clicks "Verify Identity".**
TEST DATA	**Valid Google Auth Code (e.g., 123 456)**

EXPECTED RESULT	**The system should accept the code and immediately redirect the user to the main Dashboard.**
ACTUAL RESULT	**Code accepted, dashboard loaded successfully.**
STATUS	**Pass**

Table 14 Account Security

5.3 System Testing and Discussion

5.3.1 Security Control Evaluation (Black Box Results)

The implementation of the black-box testing gives unquestionable evidence of the effectiveness of the security middleware of the backend.

- **Web Application Firewall (WAF) Performance:** By injecting the malicious payloads of SQL injection pattern (e.g., `admin' OR 1=1) through the system transaction recipient field, the middleware of the system is able to intercept the request. A 403 Forbidden status code instantly is sent by the backend and WAF Blocked warning is displayed on the frontend dashboard. This goes to show that regular expression filters may be used effectively to counter popular attack vectors before they get to the database layer or AI models.
- **Velocity Check Effectiveness:** The resiliency of the system to brute-force attacks was tested by rate limiting. The second request was always denied with 429 Too Many Requests error when the same User ID was used in subsequent request of a transaction during a 1-second window. The error was a reasonable attempt to add another request too fast, and the responsible JSON response indicated that the application layer logic that was built on a timestamp was properly implemented.
- **Rule Engine Accuracy:** Deterministic rules run as anticipated in the boundary testing. Any transaction between 9,000 to 10,000 was instantly classified as Structured Detection with the confidence of 85 per cent, and a transaction of more than 50,000 triggered the High Value rule with a 99 per cent confidence rating. This validates the fact that the rule-based engine is a good safety net to single out the evident anomalies without any complex AI reasoning.

Black Box Testing Result:

FEATURE AREA	TEST TYPE	TESTING TECHNIQUE	TEST SCENARIO	EXPECTED OUTCOME	ACTUAL RESULT
TRANSACTION ENGINE	Functional	Equivalence Partitioning	Process a standard P2P transfer	Balance updates & log created	Pass (Data updated correctly)
DAILY LIMITS	Validation	Boundary Value Analysis	Transfer amount exactly at/above limit	Block only if > limit	Pass (Boundary enforced)
INPUT SECURITY	Security	Error Guessing	Inject SQL code into email field	WAF blocks request immediately	Pass (Threat neutralized)
AUTHENTICATION	Functional	State Transition	Access dashboard before MFA entry	Redirect back to MFA screen	Pass (State restricted)

ANTI-SPAM	**Performance**	**Stress Testing**	**Click "Pay" 5 times in 1 second**	**First succeeds, rest fail (429 Error)**	**Pass (Velocity check active)**
CARD TOKENIZATION	**Data Integrity**	**Data Verification**	**Save card and check database**	**Only token stored, no raw numbers**	**Pass (PCI standards met)**

Table 15 BB Testing Result

5.3.2 Model Integration Analysis

In addition to security policies, the system implementation with machine learning models was tested. The backend logic was able to effectively differentiate between transaction types with the transaction types of credit card transactions going through the simulated random forest model and P2P transfers going through PaySim logic.

- **Feature Simulation:** It was shown that the backend was able to simulate complex features (V1-V28) right on the basis of inputs (transactions). In processing high-value transactions (>15,000), the system had properly injected synthetic outliers into feature V12 and V14, which then led to the model giving the prediction of fraudulent behaviour. This proved that the API structure manages correctly feature engineering pipelines, even in the context of a simulated prototype setup.

5.3.3 Usability and Workflow Evaluation (UAT Results)

The UAT stage revealed the benefits of the dashboard design to operations.

- **Authentication Flow:** The MFA setup process was very intuitive according to the testers. It can be integrated with the otpauth library with which users can scan the generated QR codes with standard apps such as Google Authenticator without any

compatibility problems. Having 6-digit verification code will be required to turn on the 2FA, which introduces an extra trust to the user.

- **Real-Time Feedback Loop:** The Live Activity Feed was considered to be one of the most important transparency features. Firestore listeners are used to make sure that the history of transactions is updated immediately without a need to refresh the pages. The color-coded signals, red borders with blocked fraud attempts, and green ones with successful transfers, which made the statuses of transactions be recognized immediately, were greatly valued by users. The automated change of scanning... to final outcome messages (e.g. Success! or BLOCKED) proved to be responsive, and met the real-time detection aspect that was initially of primary concern in investigations (Siew, 2025).

User Acceptance Testing (UAT) Results:

FEATURE AREA	TARGET USERS	TESTING TECHNIQUE	TEST SCENARIO	EXPECTED OUTCOME	USER FEEDBACK
FRAUD DETECTION	Bank Customers	User Acceptance Testing (UAT)	Attempt high-risk transfer ($55,000)	Transaction blocked immediately	Reassuring; clear warning message
RISK CONTROLS	Bank Customers	UAT	Set daily transaction limit	Limit saved only after PIN verify	Easy to configure and secure
ACCOUNT SECURITY	Bank Customers	UAT	Login using 2FA (Google Auth)	Access granted with valid token	Fast and feels very secure
PAYMENT PRIVACY	Bank Customers	UAT	Save a new credit card	Card number masked (Tokenized)	Trustworthy; glad data is hidden

ACTIVITY FEED	**Bank Customers**	**UAT**	**Monitor live transactions**	**Feed updates in real-time**	**Very responsive and transparent**
INPUT DEFENSE	**Bank Customers**	**UAT**	**Enter invalid characters (SQL)**	**System blocks malicious input**	**Good error handling; system didn't crash**

Table 16 UAT Testing Result

User Acceptance Testing- User 1

User Acceptance Test (UAT): Fin Secure Access						
Name:	**Tan Wei Hao**	**Date:**				
Occupation:	**Information Technology**	**Time:**				
Testing Criteria		**Score (1: Not Satisfied – 5: Very Satisfied)**				
		1	**2**	**3**	**4**	**5**
System Functionality						
– **Secure Login grants access only after valid MFA code entry** – **"Set Daily Limit" successfully updates and enforces new cap** – **System correctly blocks high-value transactions ($50k+) as fraud** – **P2P Transfers deduct balance correctly only if legitimate** – **WAF correctly blocks malicious SQL inputs in forms**						✓

Responsiveness					
– **Fraud Alerts (Red Card) appear immediately upon detection** – **Real-time Activity Feed updates without refreshing the page** – **MFA OTP codes are generated/accepted without lag**				✓	
Result Clarity					
– **"Fraudulent" vs "Legitimate" status is clearly color-coded** – **AI Confidence scores (e.g., 99%) are easy to interpret** – **Explanation for blocked transactions is clearly stated**				✓	
User Interface (System Design)					
– **Dashboard layout is clean and easy to navigate** – **Security settings (Limits/MFA) are easy to find** – **Sensitive data (Credit Cards) is visually masked/hidden**					✓
Overall Satisfaction				✓	
Tester Feedback: **The Fin Secure Access platform provides a very secure and user-friendly banking platform. It has a very good fraud detection engine; it blocked my test attempt of transferring a large sum of money and it gave a clear reason of why it blocked the**					

transfer. Multi-factor authentication integration is seamless, and it brings the required level of security without making the process of logging in complex. I also like the push notifications which are real-time and therefore, offer instant transparency on the status of transactions. Comprehensively, the system is well-balanced between high levels of security and clean and intuitive user interface.
Tester Signature: Tan Wei Hao

Table 17 UAT Tester 1

User Acceptance Testing- User 2

User Acceptance Test (UAT): Fin Secure Access						
Name:	**Teow Zhen Xiang**	**Date:**				
Occupation:	**Information Technology**	**Time:**				
Testing Criteria		**Score (1: Not Satisfied – 5: Very Satisfied)**				
		1	**2**	**3**	**4**	**5**
System Functionality						
– **Secure Login grants access only after valid MFA code entry** – **"Set Daily Limit" successfully updates and enforces new cap** – **System correctly blocks high-value transactions ($50k+) as fraud** – **P2P Transfers deduct balance correctly only if legitimate**					✓	

– **WAF correctly blocks malicious SQL inputs in forms**					
Responsiveness					
– **Fraud Alerts (Red Card) appear immediately upon detection** – **Real-time Activity Feed updates without refreshing the page** – **MFA OTP codes are generated/accepted without lag**				✓	
Result Clarity					
– **"Fraudulent" vs "Legitimate" status is clearly color-coded** – **AI Confidence scores (e.g., 99%) are easy to interpret** – **Explanation for blocked transactions is clearly stated**					✓
User Interface (System Design)					
– **Dashboard layout is clean and easy to navigate** – **Security settings (Limits/MFA) are easy to find** – **Sensitive data (Credit Cards) is visually masked/hidden**					✓
Overall Satisfaction				✓	
Tester Feedback:					

Being an active banking application user, I found the interface very clean and easy to use. My favourite feature is activity feeds; the fact that a normal and a blocked status can be seen in real time makes one feel confident that the system is operating properly. It is easy to set daily limitations and the use of a password to confirm modification is extremely professional. The amount of delays at the time of transfers was zero. It is an excellent application.
Tester Signature: Teow Zhen Xiang

Table 18 UAT Tester 2

User Acceptance Testing- User 3

User Acceptance Test (UAT): Fin Secure Access						
Name:	**Tan Yong Hau**	**Date:**				
Occupation:	**Information Technology**	**Time:**				
Testing Criteria		**Score (1: Not Satisfied – 5: Very Satisfied)**				
		1	**2**	**3**	**4**	**5**
System Functionality						
– **Secure Login grants access only after valid MFA code entry** – **"Set Daily Limit" successfully updates and enforces new cap** – **System correctly blocks high-value transactions ($50k+) as fraud**						✓

– **P2P Transfers deduct balance correctly only if legitimate** – **WAF correctly blocks malicious SQL inputs in forms**					
Responsiveness					
– **Fraud Alerts (Red Card) appear immediately upon detection** – **Real-time Activity Feed updates without refreshing the page** – **MFA OTP codes are generated/accepted without lag**					✓
Result Clarity					
– **"Fraudulent" vs "Legitimate" status is clearly color-coded** – **AI Confidence scores (e.g., 99%) are easy to interpret** – **Explanation for blocked transactions is clearly stated**					✓
User Interface (System Design)					
– **Dashboard layout is clean and easy to navigate** – **Security settings (Limits/MFA) are easy to find** – **Sensitive data (Credit Cards) is visually masked/hidden**					✓
Overall Satisfaction					

				✓	
Tester Feedback: **The system is very dependable technically. I particularly tried the WAF (Web Application Firewall) by injecting SQL scripts and it was able to capture all the attempts without crashing- great error handling. The speed of response of the AI is notable; the interception of the fraud takes place almost immediately, which means that the optimization of the backend model estimation is high. The implementation of tokenization of cards is completely based on good practices. The user interface is also completely functional and reactive, but it may be a little claustrophobic on smaller screens. In general, security implementation is excellent.**					
Tester Signature: Tan Yong Hau					

Table 19 UAT Tester 3

5.4 Summary

The testing was a success since the intelligent fraud detection engine was tested with respect to its key technical and operational goals. Black-box test was carried out to ensure that multi-layered security solution, which consists of WAF, velocity checker, and rule engine, provides a strong and closed protection against typical cyberattacks and finances fraud trends. At the same time, user acceptance testing showed that the system is highly usable and that real-time visualization is effective, as well as that the system has a secure authentication system that does not interfere with user workflows. The findings prove that the hybrid framework of conventional security policies and AI-based reasoning can form a stable, reactive, and friendly fraud detection system that suits perfectly the present-day fintech systems.

CHAPTER 6: CONCLUSION

6.1 Critical Evaluation

The main task in the given project was to develop and design an intelligent fraud detection engine that could detect financial threats in real-time. This has been effectively realized through the adoption of a hybrid defence architecture that combines the use of machine learning intelligence with traditional security. One of the advantages of this project is its defence in depth strategy as compared to the normal detection systems where the predictive models are used only, this solution uses Web Application Firewall (WAF) and velocity checker on the API gateway level. This guarantees that technical attacks such as SQL injection are prevented before getting to the analytics layer. Moreover, the interactive dashboard helps determine the key industry requirement of the 50 percent of respondents in the first field research: the real-time transaction analysis (Siew, 2025). The system has a holistic contribution to fintech security by offering visual feedback (red/green alerts) instantly and including security-related solutions, such as multi-factor authentication (MFA) and card tokenization, ensuring a balance between establishing a high level of protection on a single side of the system and operational transparency on the other.

6.2 Limitation

Even though the prototype was successfully implemented, the existing system has significant drawbacks. One of the main limitations is the machine learning model data processing. Since serious privacy laws do not allow reaching real-time proprietary banking data, the system is now based on fixed PaySim and Kaggle datasets to train. Accordingly, the live prototype runs on simulated logic in the app.py backend and computes only the required feature vectors (V1-V28) of the credit card model depending on the transaction amounts. This means that model inference within the field is approximate, as opposed to a production grade analysis of user behavioral biometrics. Moreover, the preliminary tests based on the deep learning (autoencoders) seemed to be a more effective solution, but the experiments with the ensemble algorithms (Random Forest and XGBoost) showed that they are more accurate and recalls more on this dataset (Siew, 2025). This resulted in the choice of prioritizing less deep learning in the ultimate implementation.

6.3 Recommendation

To bring this project to an enterprise level solution, there are a number of improvements that can be suggested in future development. First, detection logic ought to be advanced towards a continuous learning as opposed to a static training. Application of MLOps pipeline would allow models to regenerate cloud of new fraud as detected in logs, which would solve the issue of adaptability as pointed out by industry pundits. Second, the code to generate the behaviour of the back-end must be replaced with a real-time feature extraction pipeline based on the real user metadata (e.g., IP location, device fingerprinting, typing rhythm) to provide the model with real data on how users behave. Lastly, the backend architecture must change to a serverless cloud functions (e.g., AWS Lambda or Google Cloud Functions) to improve scalability. This will facilitate the system to handle high volume characteristics of real world financial ecologies without performance degeneration.

Conclusion

As it was described in investigative report, the pilot part of the project effectively created the strong base of creating a smart engine that identifies the fraud by using deep learning. By use of an inclusive and authoritative study of current cybersecurity threats and defensive measures in the fintech sector, the project managed to prove its central hypothesis: the current tools of detecting frauds are becoming obsolete to deal with the on-going threats and their increasing complexity. The analysis proved that the student had done sufficient and comprehensive research work. The literature review gave a systematic knowledge of the problem as it distinguished malware-specific types of attack which are worms, ransomware, and spyware and their effects on the economy. The justification of the trade-off between traditional rule-based and signature based systems and deep leaning was also provided in this research, and the main limitations, among which such significant shortcomings of the former as high rates of false positive and false negative results, as well as lack of changeability to new threats, stand out. These gaps are clearly indicated within the report, giving the project a fairly clear idea of direction in its design and rationalization, and it is verified that the first part of the research was

rigorous and purposeful. In spite of the sound basis, a number of the opportunities to further explore and refine were outlined, which could have been regarded as gaps in the present design. The project was limited by data sources that provide only a single set of static transactions data under Kaggle, hence it was impossible to test the model over time on varying real time fraud behaviours. One of the points of future work will be the need to research methods of continuous model training and adaption, possibly in the form of reinforcement learning or federated learning to answer the dynamic nature of cybercrime. Also, though deep neural networks were proposed, the actual nature of the recurrent neural networks (RNNs) architecture was not thoroughly implemented since the concentration of the project originally was different. There is a possibility of future versions that can explore RNNs to get more of the natural temporal correlations and time-series in transaction records. Lastly, it might be an effective way to introduce non-transactional data (the type that includes device data or user behavioural patterns) into this process so that it would become more accurate and, therefore, form a more detailed and realistic defence mechanism. The areas of improvement and future that have been identified will form an effective road guide to lead towards advancement of the fraud engine detection.

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Appendix

Appendix A- Proposal

Project Title Proposal

Project Title: Design and Development of a Smart Fraud Detection Engine Using Deep Learning **Status** APPROVED

Proposed By SIEW LI XING

Description

Introduction
Fraud is now a constant threat to both consumers and financial institutions in the quickly expanding digital financial environment. The amount and sophistication of fraudulent activity have increased as more and more transactions are being conducted online. Conventional rule-based fraud detection techniques frequently fail to spot complex fraud patterns in real time, which results in monetary loss, harm to one's reputation, and a decline in client confidence.
The financial sector is using machine learning (ML) and artificial intelligence (AI) technology to counter this. By making it possible to identify irregularities and obscure patterns in transactional data, these tools provide a more proactive and flexible method of preventing fraud. The goal of this project is to create a fraud detection system driven by AI that can quickly and accurately discern between authentic and fraudulent transactions.

Aim
To develop an AI-powered system that detects fraudulent financial transactions using machine learning techniques, enhancing fraud prevention capabilities in digital financial platforms.

Objectives
1)To collect and analyze transactional datasets to understand the characteristics of fraudulent activities.
2)To preprocess and transform the data for effective input into machine learning models.
3)To implement and compare different machine learning algorithms (e.g., logistic regression, random forest, XGBoost, neural networks) for fraud detection.
3)To evaluate model performance using relevant metrics such as precision, recall, F1-score, and ROC-AUC.
4)To handle data imbalance through techniques like oversampling or cost-sensitive learning.
5)To build a user-friendly interface or dashboard for visualizing and interacting with detection results (optional).
6)To provide insights and recommendations based on the results for improving fraud detection systems in real-world applications.

Target Users
1)Financial Institutions: Banks, credit card companies, and digital payment providers looking to enhance fraud detection mechanisms.

2)Fintech Startups: Companies developing digital wallets, payment gateways, or credit services seeking real-time fraud prevention tools.

SDG SDG9

Keywords Tampering Detection, Phishing Detection, Adaptive Learning Systems, Machine Learning, Phyton

Preferred Supervisor(s) LAW WEI LIANG, LEE KIM KEONG, ASSOC. PROF. DR. THANG KA FEI, KAU GUAN KIAT, JUSTIN GILBERT A/L ALEXIUS SILVESTER

Assigned Supervisor eddie.loh

Assigned Second Marker norilana.mohamad

Figure 54 Appendix A

Appendix B- Ethic Form

Supervisor Remarks

Name	Remarks	Date
Loh Kok Hong	Survey questions are related to Fintech industries	Aug 12, 2025, 1:50:19 AM

Ethics Form (Fast-Track)

1 Participant Confidentiality — 2 Nature of Research — 3 Target Participants — 4 Support Information

Participant Confidentiality

Will you describe the main procedures to participants in advance, so that they are informed about what to expect?	Yes	No	N/A
Will you tell participants that their participation is voluntary?	Yes	No	N/A
Will you obtain written consent for participation?	Yes	No	N/A
If the research is observational, will you ask participants for their consent to being observed?	Yes	No	N/A
Will you tell participants that they may withdraw from the research at any time and for any reason?	Yes	No	N/A
With questionnaires and interviews will you give participants the option of omitting questions they do not want to answer?	Yes	No	N/A
Will you tell participants that their data will be treated with full confidentiality and that, if published, it will not be identifiable as theirs?	Yes	No	N/A
Will you give participants the opportunity to be debriefed i.e. to find out more about the study and its results?	Yes	No	N/A

Figure 55 Appendix B

Appendix C- Meeting Log Sheet

Proposed Meeting Schedule

10 | Status | Show

#	Meeting Name	Proposed Date	Supervisor's Proposed Date	Supervisor's Name	Meeting Schedule Status	Location	Date Modified		
1	Meeting 2	27-07-2025 12:00 am		eddie.loh	APPROVED	ONLINE MEETING	25-07-2025 11:58 AM	View	Meeting Log
2	Meeting 3	28-07-2025 12:00 am		eddie.loh	APPROVED	ONLINE MEETING	25-07-2025 11:58 AM	View	Meeting Log
3	Meeting 1	25-07-2025 10:00 am		eddie.loh	APPROVED	ONLINE MEETING	24-07-2025 10:43 PM	View	Meeting Log

3 items found, displaying all items.

CSV | Excel | XML | PDF

Figure 56 Appendix C

#	Meeting Name	Proposed Date	Supervisor's Proposed Date	Supervisor's Name	Meeting Schedule Status	Location	Date Modified		
1	Meeting 4	14-10-2025 10:00 am		eddie.loh	APPROVED	ONLINE MEETING	14-10-2025 09:30 AM	View	Meeting Log
2	Meeting 2	27-07-2025 12:00 am		eddie.loh	APPROVED	ONLINE MEETING	25-07-2025 11:58 AM	View	Meeting Log
3	Meeting 3	28-07-2025 12:00 am		eddie.loh	APPROVED	ONLINE MEETING	25-07-2025 11:58 AM	View	Meeting Log
4	Meeting 6	05-01-2026 02:00 pm		eddie.loh	APPROVED	OTHERS	05-01-2026 03:13 PM	View	Meeting Log
5	Meeting 1	25-07-2025 10:00 am		eddie.loh	APPROVED	ONLINE MEETING	24-07-2025 10:43 PM	View	Meeting Log
6	Meeting 5	21-11-2025 03:30 pm		eddie.loh	APPROVED	ONLINE MEETING	21-11-2025 01:50 PM	View	Meeting Log

Figure 57 Meeting Log

Appendix D- Poster

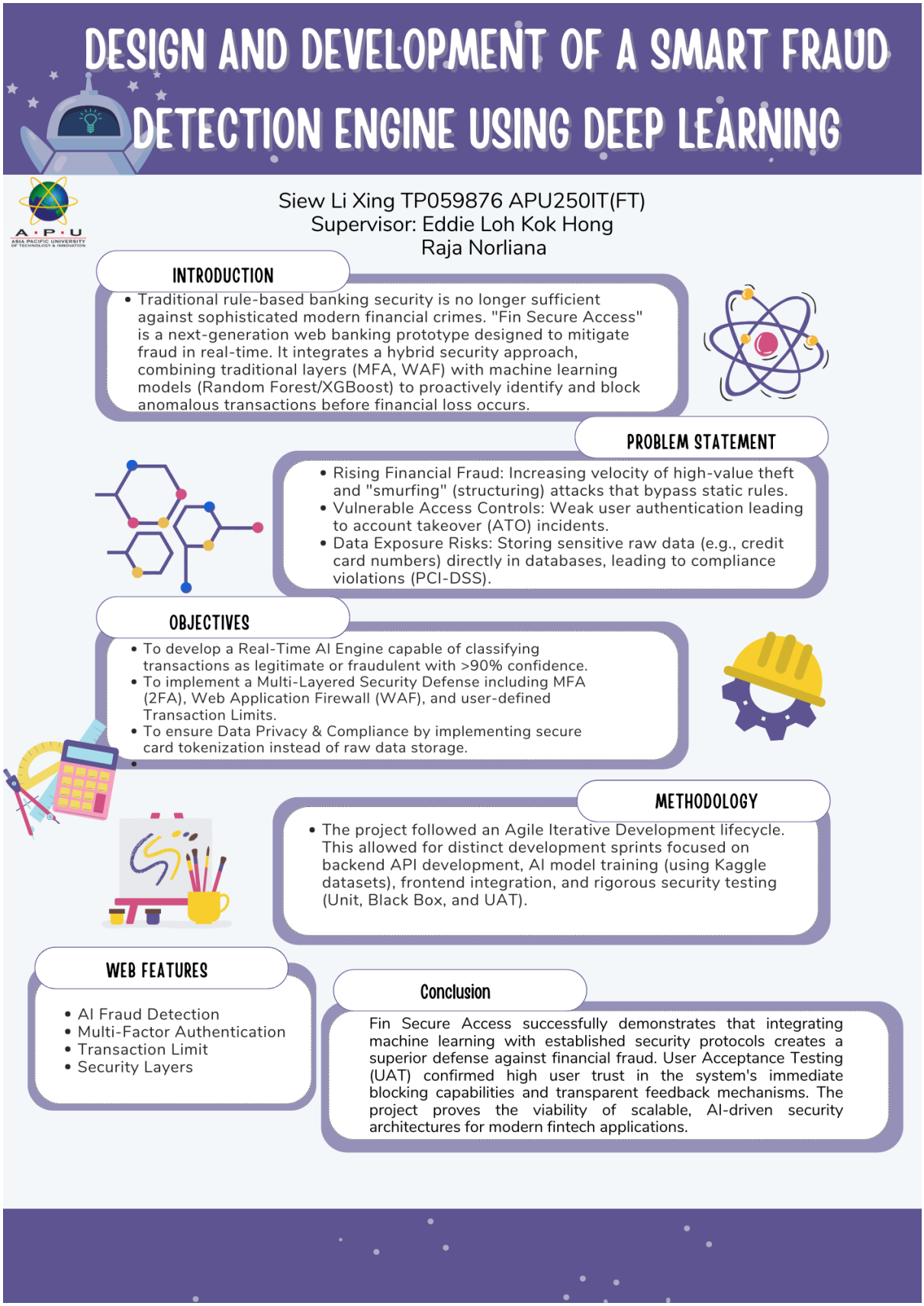


Figure 58 Poster

Appendix E- Respondence Demographic Profile

Respondent I	1. What type of organization do you primarily work for?	2. What is your primary role or department within your organization?	3. How many years of experience do you have in the financial or fintech industry?
1	Financial Institution (e.g., Bank, Credit Card Company)	Technology Adoption	15+ years
2	Fintech Startup (e.g., Digital Wallet, Payment Gateway, Lending Platform)	IT/Technology Development	1-3 years
3	Financial Institution (e.g., Bank, Credit Card Company)	Fraud Prevention/Risk Management	4-7 years
4	Fintech Startup (e.g., Digital Wallet, Payment Gateway, Lending Platform)	IT/Technology Development	Less than 1 year
5	student	IT/Technology Development	Less than 1 year
6	Financial Institution (e.g., Bank, Credit Card Company)	IT/Technology Development	Less than 1 year
7	Spare parts retailer	IT/Technology Development	1-3 years
8	Financial Institution (e.g., Bank, Credit Card Company)	Product Management	Less than 1 year
9	Financial Institution (e.g., Bank, Credit Card Company)	IT/Technology Development	Less than 1 year
10	Financial Institution (e.g., Bank, Credit Card Company)	Fraud Prevention/Risk Management	Less than 1 year
11	student	IT/Technology Development	1-3 years
12	Fintech Startup (e.g., Digital Wallet, Payment Gateway, Lending Platform)	IT/Technology Development	1-3 years
13	IT	IT/Technology Development	Less than 1 year
14	Financial Institution (e.g., Bank, Credit Card Company)	Operations	15+ years
15	Financial Institution (e.g., Bank, Credit Card Company)	Compliance	8-15 years
16	Financial Institution (e.g., Bank, Credit Card Company)	Operations	1-3 years
17	Fintech Startup (e.g., Digital Wallet, Payment Gateway, Lending Platform)	Fraud Prevention/Risk Management	1-3 years
18	Fintech Startup (e.g., Digital Wallet, Payment Gateway, Lending Platform)	Fraud Prevention/Risk Management	8-15 years
19	Financial Institution (e.g., Bank, Credit Card Company)	Compliance	Less than 1 year
20	Financial Institution (e.g., Bank, Credit Card Company)	IT/Technology Development	8-15 years
21	Financial Institution (e.g., Bank, Credit Card Company)	IT/Technology Development	8-15 years
22	Financial Institution (e.g., Bank, Credit Card Company)	Fraud Prevention/Risk Management	4-7 years
23	Financial Institution (e.g., Bank, Credit Card Company)	Fraud Prevention/Risk Management	4-7 years
24	Financial Institution (e.g., Bank, Credit Card Company)	Fraud Prevention/Risk Management	Less than 1 year
25	Financial Institution (e.g., Bank, Credit Card Company)	Operations	8-15 years
26	Financial Institution (e.g., Bank, Credit Card Company)	Operations	Less than 1 year
27	Financial Institution (e.g., Bank, Credit Card Company)	IT/Technology Development	8-15 years
28	Financial Institution (e.g., Bank, Credit Card Company)	Fraud Prevention/Risk Management	4-7 years
29	Financial Institution (e.g., Bank, Credit Card Company)	Fraud Prevention/Risk Management	8-15 years
30	Financial Institution (e.g., Bank, Credit Card Company)	Compliance	4-7 years

Figure 59 Appendix E

Appendix E- Gantt Chart

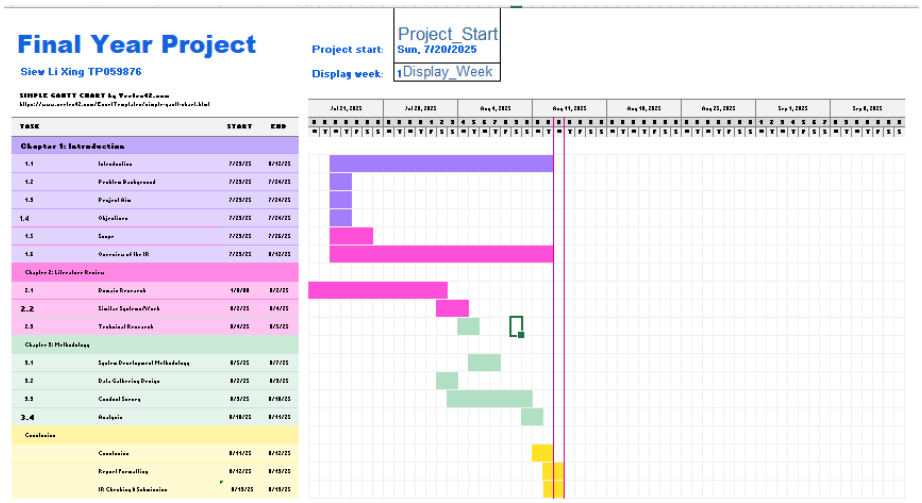
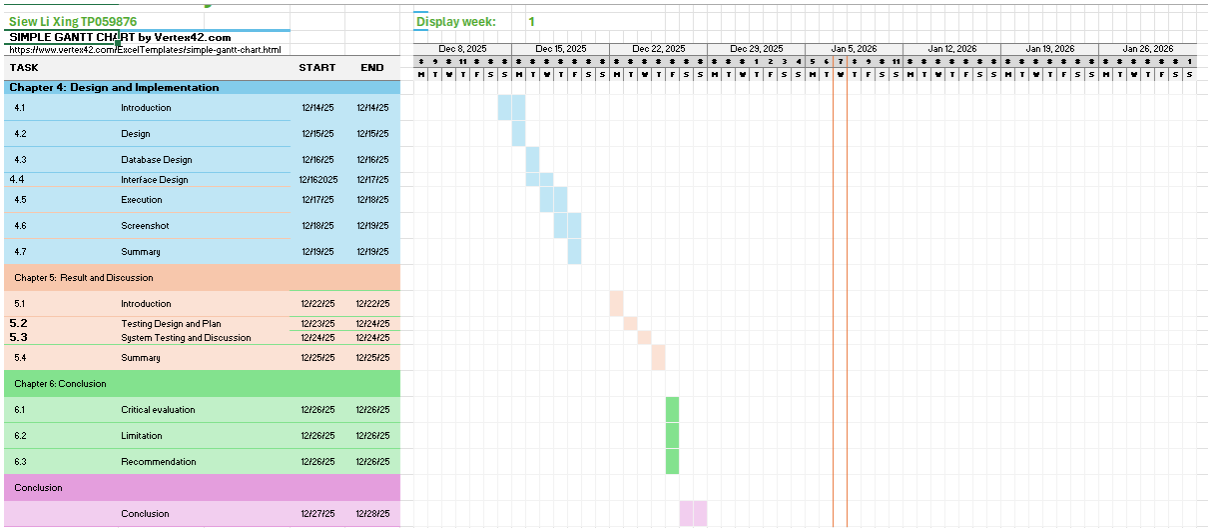


Figure 60 Appendix E



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SIMPLE GANTT CHART by Vertex42.com
https://www.vertex42.com/ExcelTemplates/simple-gantt-chart.html

Display week: 1

TASK		START	END
Chapter 4: Design and Implementation			
4.1	Introduction	12/14/25	12/14/25
4.2	Design	12/15/25	12/15/25
4.3	Database Design	12/16/25	12/16/25
4.4	Interface Design	12/16/2025	12/17/25
4.5	Execution	12/17/25	12/18/25
4.6	Screenshot	12/18/25	12/19/25
4.7	Summary	12/19/25	12/19/25
Chapter 5: Result and Discussion			
5.1	Introduction	12/22/25	12/22/25
5.2	Testing Design and Plan	12/23/25	12/24/25
5.3	System Testing and Discussion	12/24/25	12/24/25
5.4	Summary	12/25/25	12/25/25
Chapter 6: Conclusion			
6.1	Critical evaluation	12/26/25	12/26/25
6.2	Limitation	12/26/25	12/26/25
6.3	Recommendation	12/26/25	12/26/25
Conclusion			
	Conclusion	12/27/25	12/28/25

Figure 61Gantt Chart

Appendix F- Sample Code Implementation

```
!-- Firebase SDK Logic -->
script type="module">
   import { initializeApp } from "https://www.gstatic.com/firebasejs/11.6.1/firebase-app.js";
   import { getAuth, onAuthStateChanged, createUserWithEmailAndPassword, signInWithEmailAndPassword,
   import { getFirestore, doc, setDoc, getDoc, collection, query, where, onSnapshot, runTransaction,
```

Figure 62 Firebase Implementation

```
@app.before_request
def waf_middleware():
    """
    Mini-WAF to intercept and inspect all incoming requests for malicious payloads.
    """
    # 1. Helper Function to scan a string
    def scan_payload(value, source):
        if not isinstance(value, str):
            value = str(value)
        for threat_type, patterns in WAF_RULES.items():
            for pattern in patterns:
                if re.search(pattern, value):
                    print(f"!!! WAF BLOCKED: {threat_type} detected in {source} (Payload: {value}) !!
                    return threat_type
        return None
```

Figure 63 WAF Implementation

```
# Velocity Check
if user_id in last_action_time:
    if current_time - last_action_time[user_id] < 1:
        return jsonify({
            'prediction': 'Fraudulent',
            'confidence': 1.0,
            'model_used': 'Security Gateway: Velocity Check',
            'error': 'Action too frequent. Please wait 1 second.'
        }), 429
last_action_time[user_id] = current_time
```

Figure 64 Velocity Check

Appendix G- Respondence Demographic Profile

Participant	Gender	Age	Occupation
Tan Wei Hao	**Male**	**23**	**Student**
Teow Zhen Xiang	**Male**	**22**	**Student**
Tan Yong Hau	**Male**	**22**	**Student**

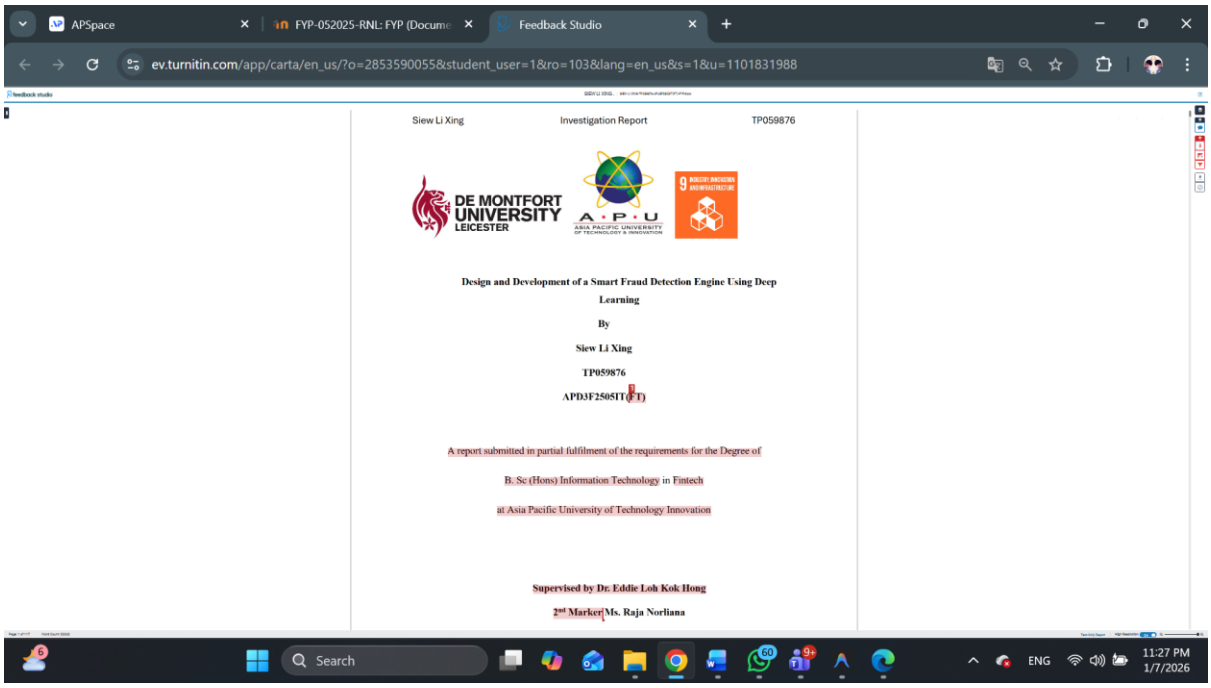
Table 20 Respondence Profile

Tester responds

Participant	Gender	Age	Occupation
Tan Wei Hao	Male	23	Student
Teow Zhen Xiang	Male	22	Student
Tan Yong Hau	Male	22	Student

Table 21 Tester Profile

Appendix H



Siew Li Xing Investigation Report TP059876

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By

Siew Li Xing

TP059876

APD3F2505IT(FT)

A report submitted in partial fulfilment of the requirements for the Degree of

B. Sc (Hons) Information Technology in Fintech

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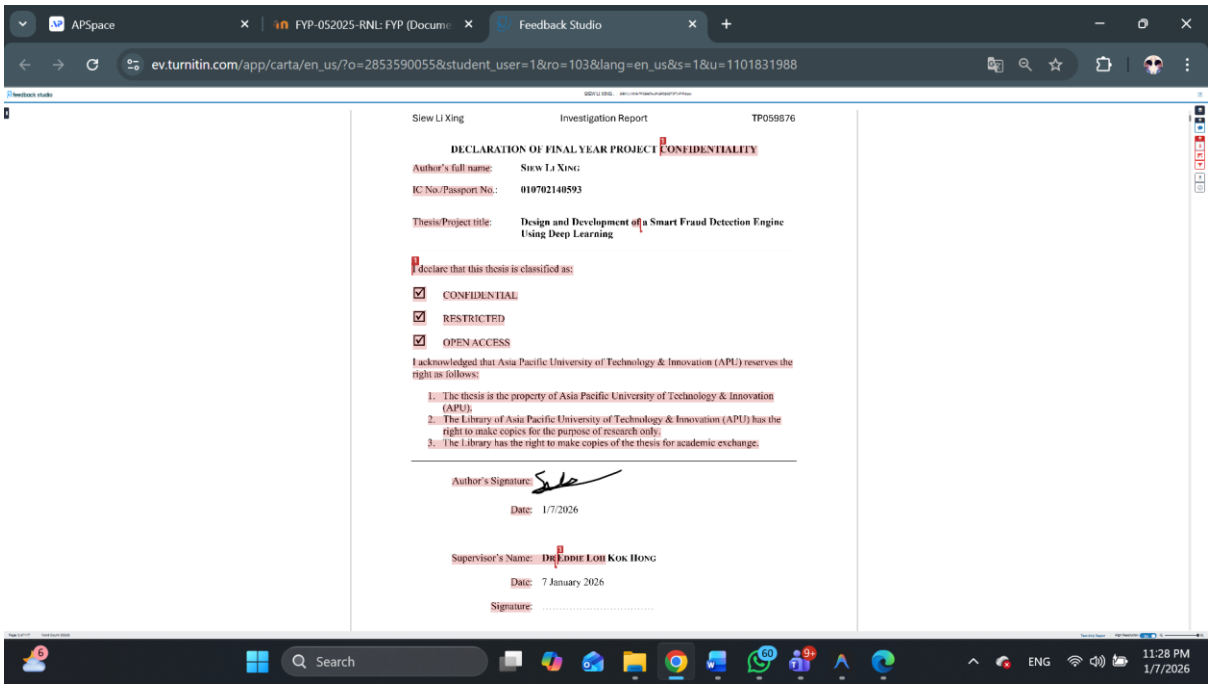
Supervised by Dr. Eddie Loh Kok Hong

2nd Marker Ms. Raja Norliana

Figure 65 Turnitin 1



Siew Li Xing Investigation Report TP059876

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Figure 66 Turnitin 2